

Asymptotic behavior of an SIQR epidemic model driven by Lévy jumps on scale-free networks

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To investigate the effect of information transmission, Lévy jumps and contact heterogeneity of individuals on the asymptotic behavior of epidemic, a stochastic SIQR epidemic model with non-monotone incidence rate and Lévy jumps on scale-free networks is constructed. At first, the global dynamics of the deterministic model is studied by constructing appropriate Lyapunov functions. Then the stochastic model is made in accordance with the ecological significance, the existence and uniqueness of the global positive solution of the stochastic SIQR model is manifested. Next, by constructing suitable stochastic Lyapunov functions and applying Itô formula with jump, the asymptotic behavior of solutions of stochastic model around equilibrium of the corresponding deterministic model is checked. At last, the correctness of the analytical results is verified by numerical simulations.

Keywords: Lévy jumps; asymptotic behavior; non-monotone incidence rate; scale-free networks; SIQR.

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1. Introduction

Examining on the transmission mechanism, pathogenesis and prevention strategies of an epidemic is an important issue concerning human health, national economy and people's livelihood. With the development of complex network research in recent years, it is found that human contact network exhibits characteristics consistent with scale-free networks [1], where the probability $P(k)$ for any individual k with links to other individuals is distributed following the power law $P(k) = Ck^{-\gamma}$ ($2 < \gamma \leq 3$). That has captured an increasing amount of interest in establishing epidemic model on scale-free networks based on traditional compartment models [2–5], which makes up for the deficiency of uniform mixing assumption of traditional compartment models to a certain extent. Li *et al.* [6] found that the network topology and isolation policy play a significant role in the epidemic threshold by investigating an SIQRS epidemic model on scale-free networks. On the basis of [6], Huang *et al.* [7] considered a network-based SIQRS epidemic model with demographics and vaccination, and the research conclusions manifest that isolate is more effective than vaccination in controlling the disease. Recently, the conclusion in [8] also shows that isolation policy plays an important role in controlling infectious diseases. In addition, at the beginning of the spread of infectious diseases, due to lack of effective medicine and isolation policy would cost a great quantity of manpower and material resource, therefore, it is necessary to take non-pharmaceutical control measures (media coverage, etc.) at this time. Inspired by the work [9], Li described the impact of information transmission on disease spreading by introducing non-monotone incidence rate (*i.e.*, $g(I)S = kIS/(1 + bI^2)$), and studied the effect of the non-monotone incidence rate on the network-based SIS epidemic model [10]. In order to comprehensively explore the influence of contact heterogeneity of individuals, information transmission, quarantine measures, demographics and vaccination, suppose that every individual has one of the four epidemiological states: susceptible (S), infected (I), quarantined (Q) and removed (R). The mechanism of disease transmission is shown in Fig. 1.

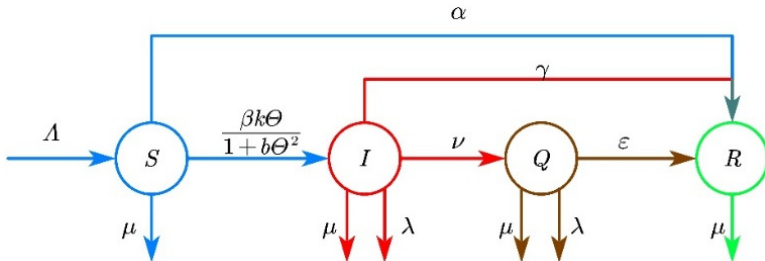


Fig. 1. The flow diagram of the SIQR epidemic model with non-monotone incidence rate.

By establishing a balance equation for the population change rate of each compartment, we obtain an SIQR epidemic model as follows:

$$\begin{cases} \frac{dS_k(t)}{dt} = -\beta k S_k(t) \frac{\Theta(t)}{1+b\Theta^2(t)} - \alpha S_k(t) - \mu S_k(t) + \Lambda, \\ \frac{dI_k(t)}{dt} = \beta k S_k(t) \frac{\Theta(t)}{1+b\Theta^2(t)} - \nu I_k(t) - \gamma I_k(t) - \mu I_k(t) - \lambda I_k(t), \\ \frac{dQ_k(t)}{dt} = \nu I_k(t) - \varepsilon Q_k(t) - \mu Q_k(t) - \lambda Q_k(t), \\ \frac{dR_k(t)}{dt} = \alpha S_k(t) + \gamma I_k(t) + \varepsilon Q_k(t) - \mu R_k(t), \end{cases} \quad (1)$$

where $S_k(t)$, $I_k(t)$, $Q_k(t)$ and $R_k(t)$ represent the density of individuals of connectivity k at time t , respectively. The parameters $b, \Lambda, \beta, \alpha, \mu, \nu, \gamma, \lambda$ and ε are considered to be positive constants. During the outbreak of an epidemic disease, a susceptible individual is infected with probability β , or vaccinated with probability α . Then an infected individual is cured with probability γ or quarantined with probability ν . The quarantined individuals are treated and become recovered individuals with probability ε with effective medical care. Furthermore, Λ represents recruitment rate of the population, μ and λ represent the natural death rate and disease-related death rate, respectively, b is the degree parameter that people take preventive and control measures. The formula $\Theta(t) = \frac{1}{\langle k \rangle} \sum_{k=1}^n k P(k) I_k(t)$ represents the probability that an susceptible individual connects to the infected individuals, where $\langle k \rangle = \sum_{k=1}^n k P(k)$ is the average degree, $P(k)$ is the degree distribution and the positive integer n is the maximum degree of the network. Using a method similar to [6], we obtain that the basic reproduction number of model (1) is

$$R_0 = \frac{\langle k^2 \rangle}{\langle k \rangle} \frac{\beta \Lambda}{(\mu + \alpha)(\nu + \gamma + \lambda + \mu)},$$

it represents the average number of secondary infectious infected by an individual of infective during whose whole course of disease in a group that all people are susceptible [15]. The basic reproduction number determines whether the disease will eventually break out. We can explain the formula for R_0 as follows, for the primary case, the average period of infection is $\frac{1}{\nu + \gamma + \lambda + \mu}$, during this period, the average number of susceptible infected by the primary case is $\beta \frac{\langle k^2 \rangle}{\langle k \rangle} \frac{\Lambda}{\mu + \alpha}$. Accordingly, the basic reproduction number R_0 has biological significance.

There always exists a disease-free equilibrium $E^0 = \{\frac{\Lambda}{\mu + \alpha}, 0, 0, \frac{\alpha \Lambda}{\mu(\mu + \alpha)}\}_{k=1}^n$, while a unique endemic equilibrium $E^* = \{S_k^*, I_k^*, Q_k^*, R_k^*\}_{k=1}^n$ exists for $R_0 > 1$,

where

$$\begin{aligned} S_k^* &= \frac{\Lambda[1 + b(\Theta^*)^2]}{(\alpha + \mu)[1 + b(\Theta^*)^2] + \beta k \Theta^*}, \\ I_k^* &= \frac{\Lambda \beta k \Theta^*}{(\nu + \gamma + \mu + \lambda)\{(\alpha + \mu)[1 + b(\Theta^*)^2] + \beta k \Theta^*\}}, \\ Q_k^* &= \frac{\Lambda \nu \beta k \Theta^*}{(\nu + \gamma + \mu + \lambda)(\mu + \varepsilon + \lambda)\{(\alpha + \mu)[1 + b(\Theta^*)^2] + \beta k \Theta^*\}}, \\ R_k^* &= \frac{\Lambda \alpha (\nu + \gamma + \mu + \lambda)(\mu + \varepsilon + \lambda)[1 + b(\Theta^*)^2] + \Lambda[\varepsilon \nu + \gamma(\mu + \varepsilon + \lambda)]\beta k \Theta^*}{\mu(\nu + \gamma + \mu + \lambda)(\mu + \varepsilon + \lambda)\{(\alpha + \mu)[1 + b(\Theta^*)^2] + \beta k \Theta^*\}}. \end{aligned}$$

The global dynamics of model (1) will be discussed in the second part of this paper. As known to all, the spread of diseases is inevitably disturbed by some stochastic factors [11–14], scholars usually use white noise to describe environmental disturbances such as climate, temperature and so on. Zhao *et al.* [15] studied a stochastic SIQS epidemic model on scale-free networks by introducing stochastic perturbations to the infected rates, and they manifested sufficient conditions for extinction and persistence of the diseases. Liu *et al.* [16] provided the sufficient conditions of stationary distribution and extinction of a stochastic SEIR epidemic model without depending on the endemic equilibrium. Note that the stochastic differential equation (SDE) with white noise is more practical than the deterministic model, but the continuous noise (white noise) still cannot well reflect the discontinuous fluctuations (hurricane, earthquake, flood, tsunami, etc.) in reality [17–20]. However, according to Lévy–Itô decomposition theorem [21], Lévy jumps is composed of Brownian motion, independent of Poisson random measure and deterministic drift term. Therefore, Lévy jumps can better describe some abrupt and discontinuous phenomena than white noise, so it is a good choice to use Lévy jumps to depict the impact of environmental mutation on the spread of diseases. Zhang *et al.* [22] investigated a stochastic SIQS epidemic model with Lévy jumps, and found that Lévy jumps can curb the disease outbreak. Lv *et al.* [23] established sufficient conditions for persistence and existence of Nontrivial T-periodic solution of a non-autonomous stochastic SIS epidemic model. Koufi *et al.* [24] discussed a stochastic SIQR epidemic model and provided sufficient conditions for the extinction and persistence of the disease. Threshold is an important parameter to characterize the spread of infectious diseases, so Zhang *et al.* [25] investigated the threshold of a stochastic SIQS epidemic model. Liu *et al.* [26] formulated a hybrid switching SIS epidemic model by introducing a standard geometric Brownian motion, a stationary Poisson point process and a continuous time finite-state Markov chain, in addition, the dynamical behavior of the model was investigated. Furthermore, Yasin *et al.* [27] developed a numerical scheme to solve the Fitzhugh–Nagumo equation in the presence of stochastic function and simulated the proposed numerical scheme in the form of the 2D and 3D plots. Although the above works

considered the effect of discontinuous fluctuations on the dynamic behavior of infectious diseases, they ignored the contact heterogeneity of individuals.

In order to make the model more accurately reflect the dynamic behavior of infectious diseases, Lévy jumps is introduced into model (1), and a stochastic SIQR epidemic model with non-monotone incidence rate and Lévy jumps on scale-free networks is obtained as follows:

$$\left\{ \begin{array}{l} dS_k(t) = \left[-\beta k S_k(t) \frac{\Theta(t)}{1+b\Theta^2(t)} - \alpha S_k(t) - \mu S_k(t) + \Lambda \right] dt + \sigma_1 S_k(t) dB_1(t) \\ \quad + \int_Z H_1(z) S_k(t-) \tilde{N}(dt, dz), \\ dI_k(t) = \left[\beta k S_k(t) \frac{\Theta(t)}{1+b\Theta^2(t)} - \nu I_k(t) - \gamma I_k(t) - \mu I_k(t) - \lambda I_k(t) \right] dt \\ \quad + \sigma_1 I_k(t) dB_1(t) + \sigma_2 I_k(t) dB_2(t) + \int_Z H_2(z) I_k(t-) \tilde{N}(dt, dz), \\ dQ_k(t) = [\nu I_k(t) - \varepsilon Q_k(t) - \mu Q_k(t) - \lambda Q_k(t)] dt + \sigma_1 Q_k(t) dB_1(t) \\ \quad + \sigma_2 Q_k(t) dB_2(t) + \int_Z H_3(z) Q_k(t-) \tilde{N}(dt, dz), \\ dR_k(t) = [\alpha S_k(t) + \gamma I_k(t) + \varepsilon Q_k(t) - \mu R_k(t)] dt + \sigma_1 R_k(t) dB_1(t) \\ \quad + \int_Z H_4(z) R_k(t-) \tilde{N}(dt, dz), \end{array} \right. \quad (2)$$

where $B_i(t) (t \geq 0) (i = 1, 2)$ stands for independent standard Brownian motion, $\sigma_i^2 > 0 (i = 1, 2)$ stands for the intensity of the white noise, $H_i(z) > -1 (i = 1, 2, 3, 4)$ indicates the intensity of the Lévy jumps. $S_k(t-), I_k(t-), Q_k(t-)$ and $R_k(t-)$ are the left limit of $S_k(t), I_k(t), Q_k(t)$ and $R_k(t)$, respectively. Let $\tilde{N}(t, U) = N(t, U) - \hat{N}(t, U)$, where $N(t, U)$ is the Poisson random measure, $\hat{N}(t, U)$ is the compensation measure of $N(t, U)$. $H_i(z) > -1 (i = 1, 2, 3, 4)$ is the intensity measure of $N(t, U)$, which satisfies $n(dt, dz) = \pi(dz)dt$, and $\pi(dz)$ is the measure on Z , besides $Z \subset (0, \infty)$, $\pi(Z) < \infty$, and satisfies $\int_Z (|z|^2 \wedge 1) \pi(dz) < \infty$ [28], the meanings of other parameters are the same as model (1).

Reviewing the existing literature, as far as we know, there are very few studies in the literature about SIQR epidemic model with Lévy jumps on scale-free networks. Accordingly, the major contribution of this paper is as follows:

- (1) On the basis of [7, 10, 21], we present a new SIQR epidemic model with Lévy jumps and non-monotone incidence rate on scale-free networks.
- (2) Consider the effect of quarantine, vaccination, information transmission, demographics and Lévy jumps on the spread of diseases.

- (3) Study the global dynamics of the deterministic model by constructing appropriate Lyapunov functions.
- (4) Obtain the asymptotic behavior of the stochastic model by applying Itô formula with jump.

The remaining part of this paper is as follows. In Sec. 2, we investigate the global dynamical behavior of the deterministic model (1). In Sec. 3, we prove that the existence of a unique global solution, and explore the asymptotic properties of stochastic model (2) around the equilibrium of the corresponding deterministic model. In Sec. 4, the correctness of the analytical results is verified by numerical simulations. In Sec. 5, we conclude the full text and prospect of the future work.

2. Global Dynamics of the Deterministic Model

Due to the high dimension of the epidemic model on the complex network, we linearize the model (1) with the linear transformation method used in [32], and then analyze the global dynamics of the disease in a deterministic environment by constructing appropriate Lyapunov functions [10].

2.1. Positivity and boundedness of solutions

We initially manifest that the solutions of model (1) with some appropriate initial conditions, see (3), are positive and the positively invariant set is obtained.

Obviously, the initial conditions for deterministic model (1) satisfy

$$\begin{aligned} 0 < S_k(0), \quad I_k(0) \leq 1, \\ 0 \leq Q_k(0), \quad R_k(0) \leq 1, \\ \Theta(0) > 0 \quad (k = 1, 2, \dots, n). \end{aligned} \quad (3)$$

Theorem 1. *Let $(S_1(t), I_1(t), Q_1(t), R_1(t), \dots, S_n(t), I_n(t), Q_n(t), R_n(t))$ be a solution of deterministic model (1) satisfying initial conditions (3), then $0 < S_k(t)$, $I_k(t)$, $Q_k(t)$, $R_k(t) < 1$ and $0 < \Theta(t) < 1$ for all $t > 0$ and $k = 1, 2, \dots, n$.*

Proof. First, we prove that $I_k(t) < 1$ for all $k = 1, 2, \dots, n$ and $t > 0$. By using the continuity of $I_k(t)$, there is a small $\zeta > 0$ such that $I_k(t) < 1$ for $t \in (0, \zeta)$. If proposition ($I_k(t) < 1$ for all $t > 0$) does not hold, we can easily find that there is $t_0 \geq \zeta > 0$, so that $I_k(t_0) = 1$ and $I_k(t) < 1$ for $t \in (0, t_0)$. It follows from the second equation of model (1), that $I'_k(t_0) = -(\nu + \gamma + \mu + \lambda) < 0$. Accordingly, there exists $\tilde{t} \in (0, t_0)$ such that $I_k(\tilde{t}) > 1$. This leads to a contradiction, therefore $I_k(t) < 1$ for all $t > 0$. Second, we prove that $I_k(t) > 0$ for all $k = 1, 2, \dots, n$ and $t > 0$. Integrating the second equation of model (1), we obtain

$$I_k(t) = I_k(0)e^{-(\nu+\gamma+\mu+\lambda)t} + \int_0^t \beta k S_k(u) \frac{\Theta(u)}{1+b\Theta^2(u)} du. \quad (4)$$

Suppose $I_k(t) > 0$ for all $t > 0$ is not true. Then there exists a positive integer k^* and $t^* > 0$ such that $I_{k^*}(t^*) = 0$. However, from (4), we obtain that

$$I_{k^*}(t^*) = I_{k^*}(0)e^{-(\nu+\gamma+\mu+\lambda)t^*} + \int_0^{t^*} \beta_k S_{k^*}(u) \frac{\Theta(u)}{1+b\Theta^2(u)} du > 0,$$

which leads to a contradiction. Consequently, we derive that $0 < I_k(t) < 1$ for all $k = 1, 2, \dots, n$ and $t > 0$.

Correspondingly, it is easy to demonstrate that $0 < \Theta(t) < 1$ for all $t > 0$. Similarly, we can obtain that $0 < I_k(t), Q_k(t), R_k(t) < 1$ for all $t > 0$. $\square$

Next, we shall consider the positively invariant set of deterministic model (1). Define

$$\begin{aligned} \Gamma = \{ & (S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \in \mathbb{R}_+^{4n} | S_k(t) + I_k(t) + Q_k(t) \\ & + R_k(t) \leq \frac{\Lambda}{\mu}, k = 1, 2, \dots, n \}, \end{aligned} \quad (5)$$

where $\mathbb{R}_+^{4n} = \{(x_1, x_2, \dots, x_{4n}) | x_i \geq 0\}$.

Lemma 1. *The set Γ is the positively invariant set of the deterministic model (1).*

Proof. From Eq. (1), we obtain

$$\frac{dN_k(t)}{dt} = \Lambda - \mu N_k(t) - \lambda(I_k(t) + Q_k(t)) \leq \Lambda - \mu N_k(t).$$

Thus

$$N_k(t) \leq N_k(0)e^{-\mu t} + \frac{\Lambda}{\mu}.$$

It follows that $\lim_{t \rightarrow +\infty} \sup N_k(t) = \frac{\Lambda}{\mu}$, which implies that the solution of model (1) starting from $\mathbb{R}_+^{4n}$ eventually tends to or stays in a subset Γ of $\mathbb{R}_+^{4n}$. Therefore, the feasible region of model (1) is Γ , that is to say, Γ is a positive invariant set of model (1). $\square$

Next, we always presume that $(S_1(t), I_1(t), Q_1(t), R_1(t), \dots, S_n(t), I_n(t), Q_n(t), R_n(t)) \in \Gamma$.

2.2. Stability of the disease-free equilibrium

In this subsection, the globally asymptotically stable of the disease-free equilibrium will be investigated. We firstly examine the local asymptotic stability of the disease-free equilibrium.

Theorem 2. *The disease-free equilibrium E^0 of the deterministic model (1) is locally asymptotically stable for $R_0 < 1$, and it is unstable for $R_0 > 1$.*

Proof. Without losing generality, let $\tilde{S}_k = S_k - S_k^0$, $\tilde{I}_k = I_k - I_k^0$, $\tilde{Q}_k = Q_k - Q_k^0$, $\tilde{R}_k = R_k - R_k^0$, then the linear system of (1) can be written as

$$\begin{cases} \frac{d\tilde{S}_k(t)}{dt} = -\beta k S_k^0 \frac{\tilde{\Theta}(t)}{1 + b\tilde{\Theta}^2(t)} - \alpha \tilde{S}_k(t) - \mu \tilde{S}_k(t), \\ \frac{d\tilde{I}_k(t)}{dt} = \beta k S_k^0 \frac{\tilde{\Theta}(t)}{1 + b\tilde{\Theta}^2(t)} - \nu \tilde{I}_k(t) - \gamma \tilde{I}_k(t) - \mu \tilde{I}_k(t) - \lambda \tilde{I}_k(t), \\ \frac{d\tilde{Q}_k(t)}{dt} = \nu \tilde{I}_k(t) - \varepsilon \tilde{Q}_k(t) - \mu \tilde{Q}_k(t) - \lambda \tilde{Q}_k(t), \\ \frac{d\tilde{R}_k(t)}{dt} = \alpha \tilde{S}_k(t) + \gamma \tilde{I}_k(t) + \varepsilon \tilde{Q}_k(t) - \mu \tilde{R}_k(t). \end{cases} \quad (6)$$

To investigate the stability of (6), we suppose the solution for Eq. (6) with the exponential form as follows:

$$\begin{aligned} \tilde{S}_k(t) &= S_{k0} e^{\rho t}, \quad \tilde{I}_k(t) = I_{k0} e^{\rho t}, \\ \tilde{Q}_k(t) &= Q_{k0} e^{\rho t}, \quad \tilde{R}_k(t) = R_{k0} e^{\rho t}, \quad (k = 1, 2, \dots, n). \end{aligned} \quad (7)$$

where $S_{k0}, I_{k0}, Q_{k0}, R_{k0}$ represent the initial values. Substituting (7) into (6) yields

$$\begin{aligned} &\rho(S_{10}, \dots, S_{n0}, I_{10}, \dots, I_{n0}, Q_{10}, \dots, Q_{n0}, R_{10}, \dots, R_{n0})^T \\ &= \mathbf{A}(S_{10}, \dots, S_{n0}, I_{10}, \dots, I_{n0}, Q_{10}, \dots, Q_{n0}, R_{10}, \dots, R_{n0})^T, \end{aligned}$$

where ρ is the eigenvalue of the linear system Eq. (6). Let $a_k = \frac{1}{\langle k \rangle} k P(k)$. Since matrix $\mathbf{A}$ is large, we denote matrix $\mathbf{A}$ as $\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{0} \\ \mathbf{A}_2 & \mathbf{A}_3 \end{pmatrix}_{4n \times 4n}$, where

$$\mathbf{A}_1 = \begin{pmatrix} -(\alpha + \mu) & \cdots & 0 & -\beta S_1^0 a_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & -(\alpha + \mu) & -\beta n S_n^0 a_1 \\ 0 & \cdots & 0 & -(\nu + \gamma + \mu + \lambda) + \beta S_1^0 a_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & \beta n S_n^0 a_1 \\ \cdots & & -\beta S_1^0 a_n \\ \ddots & & \vdots \\ \cdots & & -\beta n S_n^0 a_n \\ \cdots & & \beta S_1^0 a_n \\ \ddots & & \vdots \\ \cdots & & -(\nu + \gamma + \mu + \lambda) + \beta n S_n^0 a_n \end{pmatrix}_{2n \times 2n},$$

$$\mathbf{A}_2 = \begin{pmatrix} 0 & \cdots & 0 & \nu & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & \nu \\ \alpha & \cdots & 0 & \gamma & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha & 0 & \cdots & \gamma \end{pmatrix}_{2n \times 2n},$$

$$\mathbf{A}_3 = \begin{pmatrix} -(\varepsilon + \mu + \lambda) & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -(\varepsilon + \mu + \lambda) & 0 & \cdots & 0 \\ \varepsilon & \cdots & 0 & -\mu & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \varepsilon & 0 & \cdots & -\mu \end{pmatrix}_{2n \times 2n}.$$

By direct calculation, there are eigenvalues equaling to $-\mu$ with multiplicity n , eigenvalues equal to $-(\varepsilon + \mu + \lambda)$ with multiplicity n , and eigenvalues equal to $-(\alpha + \mu)$ with multiplicity n . To obtain other eigenvalues of matrix $\mathbf{A}$, define

$$\mathbf{\Lambda} = \begin{pmatrix} -(\nu + \gamma + \mu + \lambda) + \beta S_1^0 a_1 & \beta S_1^0 a_2 & \cdots & \beta S_1^0 a_n \\ \beta 2S_2^0 a_1 & -(\nu + \gamma + \mu + \lambda) + \beta 2S_2^0 a_2 & \cdots & \beta 2S_2^0 a_n \\ \vdots & \vdots & \ddots & \vdots \\ \beta nS_n^0 a_1 & \beta nS_n^0 a_2 & \cdots & -(\nu + \gamma + \mu + \lambda) + \beta nS_n^0 a_n \end{pmatrix}_{n \times n}.$$

Carrying out similarity transformation, the similarity matrix of matrix $\mathbf{\Lambda}$ can be obtained as follows:

$$\mathbf{\Lambda}^* = \begin{pmatrix} -(\nu + \gamma + \mu + \lambda) & 0 & \cdots & \beta S_1^0 a_n \\ 0 & -(\nu + \gamma + \mu + \lambda) & \cdots & \beta(S_1^0 a_1 + 2S_2^0 a_2) \frac{a_n}{a_2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -(\nu + \gamma + \mu + \lambda) + \beta \sum_{k=1}^n k S_k^0 a_k \end{pmatrix}_{n \times n}.$$

We can easily obtain that there is an eigenvalue equal to $-(\nu + \gamma + \mu + \lambda)$ with multiplicity $n - 1$, and the n th eigenvalue is

$$\begin{aligned}\eta &= -(\nu + \gamma + \mu + \lambda) + \beta \sum_{k=1}^n k S_k^0 a_k \\ &= (\nu + \gamma + \mu + \lambda) \left[\frac{\beta \sum_{k=1}^n k P(k) \cdot k \cdot \Lambda}{(\nu + \gamma + \mu + \lambda) \cdot \langle k \rangle \cdot (\mu + \alpha)} - 1 \right] \\ &= (\nu + \gamma + \mu + \lambda) \left[\frac{\langle k^2 \rangle \beta \Lambda}{\langle k \rangle (\mu + \alpha) (\nu + \gamma + \mu + \lambda)} - 1 \right] \\ &= (\nu + \gamma + \mu + \lambda) (R_0 - 1),\end{aligned}$$

for the matrix $\mathbf{\Lambda}^*$, where $\eta < 0$ if and only if $R_0 < 1$. Accordingly, the disease-free equilibrium is local asymptotically stable when $R_0 < 1$, and it is unstable when $R_0 > 1$. $\square$

We next prove the disease-free equilibrium is globally attractive.

Theorem 3. *If $R_0 < 1$, then the disease-free equilibrium E^0 of the deterministic model (1) is globally asymptotically stable.*

Proof. We first prove that $\lim_{t \rightarrow +\infty} I_k(t) = 0$. From Eq. (1), we can easily see that

$$\frac{dS_k(t)}{dt} \leq \Lambda - (\mu + \alpha) S_k(t).$$

By [29], we have

$$\lim_{t \rightarrow +\infty} \sup S_k(t) \leq \frac{\Lambda}{\mu + \alpha}.$$

Therefore, for any enough small $\varepsilon_1 > 0$, there exists $t_1 > 0$ such that $S_k(t) \leq \frac{\Lambda}{\mu + \alpha} + \varepsilon_1$ for all $t > t_1$. It follows that

$$\frac{dI_k(t)}{dt} \leq \beta k \left(\frac{\Lambda}{\mu + \alpha} + \varepsilon_1 \right) \frac{\Theta(t)}{1 + b\Theta^2(t)} - (\nu + \gamma + \mu + \lambda) I_k(t).$$

Next, we construct a homologous system satisfying $M_k(0) = I_k(0) \geq 0$ as follows:

$$\frac{dM_k(t)}{dt} = \beta k \left(\frac{\Lambda}{\mu + \alpha} + \varepsilon_1 \right) \frac{\tilde{\Theta}(t)}{1 + b\tilde{\Theta}^2(t)} - (\nu + \gamma + \mu + \lambda) M_k(t), \quad (8)$$

where $\tilde{\Theta}(t) = \frac{1}{\langle k \rangle} \sum_{k=1}^n kP(k)M_k(t)$.

Now, we construct the Lyapunov function $V(t) = \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle(\nu + \gamma + \mu + \lambda)} M_k(t)$. Then we obtain

$$\begin{aligned} \left. \frac{dV(t)}{dt} \right|_{(8)} &= \sum_{k=1}^n \left[\frac{\beta k^2 P(k)}{\langle k \rangle(\nu + \gamma + \mu + \lambda)} \left(\frac{\Lambda}{\mu + \alpha} + \varepsilon_1 \right) \frac{\tilde{\Theta}(t)}{1 + b\tilde{\Theta}^2(t)} - \frac{kP(k)}{\langle k \rangle} M_k(t) \right] \\ &\leq \sum_{k=1}^n \left[\frac{\beta k^2 P(k)}{\langle k \rangle(\nu + \gamma + \mu + \lambda)} \left(\frac{\Lambda}{\mu + \alpha} + \varepsilon_1 \right) \tilde{\Theta}(t) - \frac{kP(k)}{\langle k \rangle} M_k(t) \right] \\ &= \tilde{\Theta}(t) \left[R_0 + \frac{\langle k^2 \rangle \beta \Lambda}{\langle k \rangle(\mu + \alpha)(\nu + \gamma + \mu + \lambda)} \varepsilon_1 - 1 \right]. \end{aligned}$$

Since $R_0 < 1$, we can choose an arbitrarily enough small $\varepsilon_1 > 0$ such that

$$R_0 + \frac{\langle k^2 \rangle \beta \Lambda}{\langle k \rangle(\mu + \alpha)(\nu + \gamma + \mu + \lambda)} \varepsilon_1 \leq 1.$$

Then we get that $\frac{dV(t)}{dt} \leq 0$ for all $M_k(t) \geq 0$, and $\frac{dV(t)}{dt} = 0$ only if $\tilde{\Theta}(t) = 0$. So $M_k(t) = 0$, and then $\lim_{t \rightarrow +\infty} M_k(t) = 0$.

In addition, $0 \leq I_k(t) \leq M_k(t)$ for all $t > 0$, by comparison theorem, we have $\lim_{t \rightarrow +\infty} I_k(t) = 0$. When $t \rightarrow +\infty$, the limit equation of model (1) is

$$\begin{cases} \frac{dS_k(t)}{dt} = \Lambda - (\mu + \alpha)S_k(t), \\ \frac{dQ_k(t)}{dt} = -(\varepsilon + \mu + \lambda)Q_k(t), \\ \frac{dR_k(t)}{dt} = \alpha S_k(t) + \varepsilon Q_k(t) - \mu R_k(t). \end{cases} \quad (9)$$

By the first and second equations of (9), we obtain

$$S_k(t) = \frac{\Lambda}{\mu + \alpha} + C_1 e^{-(\mu + \alpha)t}, \quad Q_k(t) = C_2 e^{-(\varepsilon + \mu + \lambda)t}.$$

Thus

$$\lim_{t \rightarrow +\infty} S_k(t) = \frac{\Lambda}{\mu + \alpha}, \quad \lim_{t \rightarrow +\infty} Q_k(t) = 0. \quad (10)$$

Substituting (10) into the third expression of (9) gives

$$\lim_{t \rightarrow +\infty} R_k(t) = \frac{\alpha \Lambda}{\mu(\mu + \alpha)}.$$

These demonstrate that the disease-free equilibrium E^0 of the deterministic model (1) is globally attractive if $R_0 < 1$. Combining with Theorem 2, we manifest that the

disease-free equilibrium E^0 of the deterministic model (1) is globally asymptotically stable if $R_0 < 1$. $\square$

2.3. Stability of the endemic equilibrium

Similarly, the globally asymptotical stability of the endemic equilibrium will be discussed in the following.

Theorem 4. *The endemic equilibrium E^* of the deterministic model (1) is locally asymptotically stable for $R_0 > 1$.*

Proof. Without losing generality, let $\widehat{S}_k = S_k - S_k^*$; $\widehat{I}_k = I_k - I_k^*$; $\widehat{Q}_k = Q_k - Q_k^*$; $\widehat{R}_k = R_k - R_k^*$, then the linear system of (1) can be written as

$$\begin{cases} \frac{d\widehat{S}_k(t)}{dt} = -\beta k S_k^* \frac{\widehat{\Theta}(t)}{1+b\widehat{\Theta}(t)} - \alpha \widehat{S}_k(t) - \mu \widehat{S}_k(t) - (\alpha + \mu) S_k^* + \Lambda, \\ \frac{d\widehat{I}_k(t)}{dt} = \beta k S_k^* \frac{\Theta(t)}{1+b\Theta^2(t)} - \nu \widehat{I}_k(t) - \gamma \widehat{I}_k(t) - \mu \widehat{I}_k(t) \\ \quad - \lambda \widehat{I}_k(t) - (\nu + \gamma + \mu + \lambda) I_k^*, \\ \frac{d\widehat{Q}_k(t)}{dt} = \nu \widehat{I}_k(t) - \varepsilon \widehat{Q}_k(t) - \mu \widehat{Q}_k(t) - \lambda \widehat{Q}_k(t), \\ \frac{d\widehat{R}_k(t)}{dt} = \alpha \widehat{S}_k(t) + \gamma \widehat{I}_k(t) + \varepsilon \widehat{Q}_k(t) - \mu \widehat{R}_k(t). \end{cases} \quad (11)$$

The Jacobin matrix of system (11) at the endemic equilibrium is $\mathbf{J} = \begin{pmatrix} \mathbf{J}_1 & \mathbf{0} \\ \mathbf{J}_2 & \mathbf{J}_3 \end{pmatrix}_{4n \times 4n}$, $a_k = \frac{1}{\langle k \rangle} k P(k)$. Let $\frac{1-b(\Theta^*)^2}{(1+b(\Theta^*)^2)^2} = \Omega^*$, where

$$\mathbf{J}_1 = \begin{pmatrix} -(\alpha + \mu) & \cdots & 0 & -\beta a_1 \Omega^* S_1^* \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & -(\alpha + \mu) & -\beta a_n \Omega^* S_n^* \\ 0 & \cdots & 0 & -(\nu + \gamma + \mu + \lambda) + \beta a_1 \Omega^* S_1^* \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & \beta a_n \Omega^* S_n^* \\ \cdots & & -\beta a_n \Omega^* S_1^* & \\ \ddots & & \vdots & \\ \cdots & & -\beta a_n \Omega^* S_n^* & \\ \cdots & & \beta a_n \Omega^* S_1^* & \\ \ddots & & \vdots & \\ \cdots & & -(\nu + \gamma + \mu + \lambda) + \beta a_n \Omega^* S_n^* \end{pmatrix}_{2n \times 2n},$$

$$\mathbf{J}_2 = \begin{pmatrix} 0 & \cdots & 0 & \nu & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & \nu \\ \alpha & \cdots & 0 & \gamma & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \alpha & 0 & \cdots & \gamma \end{pmatrix}_{2n \times 2n},$$

$$\mathbf{J}_3 = \begin{pmatrix} -(\varepsilon + \mu + \lambda) & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & -(\varepsilon + \mu + \lambda) & 0 & \cdots & 0 \\ \varepsilon & \cdots & 0 & -\mu & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \varepsilon & 0 & \cdots & -\mu \end{pmatrix}_{2n \times 2n}.$$

Obviously, the matrix $\mathbf{J}$ has an eigenvalue equaling to $-\mu$ with multiplicity n , an eigenvalue equaling to $-(\varepsilon + \mu + \lambda)$ with multiplicity n , an eigenvalue equaling to $-(\alpha + \mu)$ with multiplicity n . To explore other eigenvalues of matrix $\mathbf{J}$, define

$$\mathbf{T} = \begin{pmatrix} -(\nu + \gamma + \mu + \lambda) + \beta a_1 \Omega^* S_1^* & \beta a_2 \Omega^* S_1^* & & & \\ \beta 2a_1 \Omega^* S_2^* & -(\nu + \gamma + \mu + \lambda) + \beta 2a_2 \Omega^* S_2^* & & & \\ \vdots & \vdots & & & \\ \beta na_1 \Omega^* S_n^* & \beta na_2 \Omega^* S_n^* & & & \\ \cdots & \beta a_n \Omega^* S_1^* & & & \\ \cdots & \beta 2a_n \Omega^* S_2^* & & & \\ \ddots & \vdots & & & \\ \cdots & -(\nu + \gamma + \mu + \lambda) + \beta na_n \Omega^* S_n^* & & & \end{pmatrix}_{n \times n},$$

carrying out similarity transformation to the matrix $\mathbf{T}$, the similarity transformation matrix $\mathbf{T}^*$ is obtained as follows:

$$\mathbf{T}^* = \begin{pmatrix} -(\nu + \gamma + \mu + \lambda) & 0 & \cdots & \beta a_n \Omega^* S_1^* \\ 0 & -(\nu + \gamma + \mu + \lambda) & \cdots & \beta \Omega^* (S_1^* a_1 + 2S_2^* a_2) \frac{a_n}{a_2} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -(\nu + \gamma + \mu + \lambda) + \beta \Omega^* \sum_{k=1}^n k S_k^* a_k \end{pmatrix}_{n \times n}.$$

It can be seen that the matrix $\mathbf{T}^*$ has an eigenvalue equal to $-(\nu + \gamma + \mu + \lambda)$ with multiplicity $n - 1$, and the n th eigenvalue is

$$\begin{aligned} j &= -(\nu + \gamma + \mu + \lambda) + \beta \frac{1 - b(\Theta^*)^2}{(1 + b(\Theta^*)^2)^2} \sum_{k=1}^n k \frac{kP(k)}{\langle k \rangle} \\ &\quad \times \frac{\Lambda(1 + b(\Theta^*)^2)}{(\alpha + \mu)(1 + b(\Theta^*)^2) + \beta k \Theta^*} \\ &< (\nu + \gamma + \mu + \lambda) \left[-1 + \frac{1}{1 + b(\Theta^*)^2} \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle} I_k^* \frac{1}{\Theta^*} \right] \\ &= (\nu + \gamma + \mu + \lambda) \left(-1 + \frac{1}{1 + b(\Theta^*)^2} \right) \\ &< 0. \end{aligned}$$

Accordingly, the endemic equilibrium is locally asymptotically stable when $R_0 > 1$. The global attractivity of endemic equilibrium will be demonstrated by constructing suitable Lyapunov functions in the following. $\square$

Theorem 5. *If $R_0 > 1$, then the endemic equilibrium E^* of deterministic model (1) is globally asymptotically stable.*

Proof. When $R_0 > 1$, the endemic equilibrium E^* satisfies the following equation:

$$\begin{cases} -\beta k S_k^* \frac{\Theta^*}{1 + b(\Theta^*)^2} - \alpha S_k^* - \mu S_k^* + \Lambda = 0, \\ \beta k S_k^* \frac{\Theta^*}{1 + b(\Theta^*)^2} - \nu I_k^* - \gamma I_k^* - \mu I_k^* - \lambda I_k^* = 0, \\ \nu I_k^* - \varepsilon Q_k^* - \mu Q_k^* - \lambda Q_k^* = 0, \\ \alpha S_k^* + \gamma I_k^* + \varepsilon Q_k^* - \mu R_k^* = 0. \end{cases}$$

Consider the following Lyapunov function:

$$V = \frac{1}{2} \sum_{k=1}^n \xi(k) (S_k - S_k^* + I_k - I_k^*)^2 + \rho^* \left(\Theta - \Theta^* - \Theta^* \ln \frac{\Theta}{\Theta^*} \right) =: V_1 + \rho^* V_2,$$

where $\xi(k) = \frac{kP(k)}{\langle k \rangle}$, $\rho^* = \frac{(\alpha + 2\mu + \nu + \gamma + \lambda)(1 + b(\Theta^*)^2)}{\beta \langle k^2 \rangle}$. Thus

$$\begin{aligned} \frac{dV_1(t)}{dt} &= \sum_{k=1}^n \xi(k) (S_k - S_k^* + I_k - I_k^*) (\Lambda - \alpha S_k - \mu S_k - \nu I_k - \gamma I_k - \mu I_k - \lambda I_k) \\ &= \sum_{k=1}^n \xi(k) [-(\mu + \alpha)(S_k - S_k^*)^2 - (\nu + \gamma + \mu + \lambda)(I_k - I_k^*)^2] \\ &\quad - (\alpha + 2\mu + \nu + \gamma + \lambda)(S_k - S_k^*)(\Theta - \Theta^*), \end{aligned}$$

and

$$\begin{aligned}
 \frac{dV_2(t)}{dt} &= \left(1 - \frac{\Theta^*}{\Theta}\right) \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle} \left(\beta k S_k \frac{\Theta}{1 + b\Theta^2} - \nu I_k - \gamma I_k - \mu I_k - \lambda I_k \right) \\
 &= (\Theta - \Theta^*) \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle} \left(\beta k \frac{S_k}{1 + b\Theta^2} - \beta k \frac{S_k^*}{1 + b(\Theta^*)^2} \right) \\
 &= (\Theta - \Theta^*) \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle} \left(\beta k \frac{S_k}{1 + b\Theta^2} - \beta k \frac{S_k}{1 + b(\Theta^*)^2} \right. \\
 &\quad \left. + \beta k \frac{S_k}{1 + b(\Theta^*)^2} - \beta k \frac{S_k^*}{1 + b(\Theta^*)^2} \right) \\
 &= \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle} \beta k S_k \left(\frac{1}{1 + b\Theta^2} - \frac{1}{1 + b(\Theta^*)^2} \right) (\Theta - \Theta^*) \\
 &\quad + \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle} \frac{\beta k}{1 + b(\Theta^*)^2} (S_k - S_k^*) (\Theta - \Theta^*) \\
 &\leq \frac{\beta \langle k^2 \rangle}{\langle k \rangle} \frac{1}{1 + b(\Theta^*)^2} (S_k - S_k^*) (\Theta - \Theta^*).
 \end{aligned}$$

It follows that

$$\frac{dV}{dt} \leq -(\mu + \alpha) \sum_{k=1}^n \xi(k) (S_k - S_k^*)^2 - (\nu + \gamma + \mu + \lambda) \sum_{k=1}^n \xi(k) (I_k - I_k^*)^2.$$

Clearly, $\frac{dV}{dt} \leq 0$ for all $(S_k^*, I_k^*)_{k=1}^n$, and $\frac{dV}{dt} = 0$ if and only if $S_k = S_k^*, I_k = I_k^*$. Furthermore, from the third equation of model (1), we have

$$\begin{aligned}
 Q_k(t) &= \left[Q_k(0) + \int_{t_0}^t \nu I_k(\tau) \exp[(\varepsilon + \mu + \lambda)(\tau - t_0)] d\tau \right] \\
 &\quad \cdot \exp[-(\varepsilon + \mu + \lambda)(t - t_0)].
 \end{aligned}$$

By L'Hospital's rule, we obtain

$$\lim_{t \rightarrow +\infty} Q_k(t) = \lim_{t \rightarrow +\infty} \frac{\nu I_k(t)}{\varepsilon + \mu + \lambda} = \frac{\nu I_k^*}{\varepsilon + \mu + \lambda} = Q_k^*.$$

Similarly,

$$\lim_{t \rightarrow +\infty} R_k(t) = \lim_{t \rightarrow +\infty} \frac{\alpha S_k^* + \gamma I_k^* + \varepsilon Q_k^*}{\mu} = R_k^*.$$

Consequently, the endemic equilibrium E^* is globally attractive when $R_0 > 1$, and it is locally asymptotically stable from Theorem 4, so the endemic equilibrium E^* is globally asymptotically stable when $R_0 > 1$. $\square$

3. Asymptotic Behavior of the Stochastic Model

On the basis of Sec. 2, we explore the global dynamics of diseases in stochastic environment by constructing appropriate stochastic Lyapunov functions [24] and combining Itô formula with jump. We first provide a lemma that will be used later.

Lemma 2 (Itô formula with jump [30]). *Let the operator L associate with the following n -dimensional SDE, i.e.*

$$dX = \phi(t, X(t))dt + \psi(t, X(t))dB(t) + \int_Y H(t, y)\tilde{N}(dt, dy)$$

and

$$\begin{aligned} LX(t-) &= \sum_{i=1}^n \frac{\partial X(t-)}{\partial x_i} \phi_i(t, X) + \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 X(t-)}{\partial x_i \partial x_j} [\psi^T(t, X)\psi(t, X)]_{ij} \\ &\quad + \int_Y \left[X((t-) + H(t, y)) - X(t-) - \frac{\partial X(t-)}{\partial x_i} H(t, y) \right] v(dy), \end{aligned}$$

where $X = (x_1, x_2, \dots, x_n)$. If L acts on a function $F \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^n; \mathbb{R})$. Then

$$\begin{aligned} LF(X(t)) &= F_x(X(t-))\phi(t, X) + \frac{1}{2} \text{trace}(\psi^T(t, X(t-))F_{xx}\psi(t, X(t-))) \\ &\quad + \int_Y [F(X(t-) + H(t, y)) - F(X(t-)) - F_x(X(t-))H(t, y)]v(dy), \end{aligned}$$

where $F_x = (\frac{\partial F}{\partial x_1}, \frac{\partial F}{\partial x_2}, \dots, \frac{\partial F}{\partial x_n})$, $F_{xx} = (\frac{\partial^2 F}{\partial x_i \partial x_j})_{n \times n}$.

Then by Itô formula, we obtain

$$\begin{aligned} dF(X(t)) &= LF(X(t-))dt + F_x(X(t-))\psi(t, X)dB(t) \\ &\quad + \int_Y [F(X(t-) + H(t, y)) - F(X(t-))]\tilde{N}(dt, dy). \end{aligned}$$

3.1. Existence of a unique global solution

The infectious disease model has biological significance only when it has a unique global positive solution. Next, we manifest that the stochastic model (2) has a global positive solution by Theorem 6. Before discussing the existence and uniqueness of the global solution of the stochastic model (2), we make the following provisions:

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, P)$ be a complete probability space with a filtration $\{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions, $B_i(t) (i = 1, 2)$ be defined on the complete probability space. Define

$$\mathbb{R}_+^{4n} = \{(x_1, x_2, \dots, x_{4n}) | x_i \geq 0, i = 1, 2, \dots, 4n\}.$$

The Lévy jumps in stochastic model (2) satisfy conditions (A_1) and (A_2) , for the coefficients of the diffusion term, suppose that for every $h > 0$, there exists $l_h > 0$

such that

- (A₁) $\int_{\mathbb{Z}} |H_i(x, z) - H_i(y, z)|^2 \pi(dz) \leq l_h |x - y|^2, (i = 1, 2, 3, 4)$, where $H_1(x, z) = H_1(z)S_k(t-), H_2(x, z) = H_2(z)I_k(t-), H_3(x, z) = H_3(z)Q_k(t-), H_4(x, z) = H_4(z)R_k(t-), |x| \vee |y| \leq h$.
- (A₂) $\int_{\mathbb{Z}} |\log(1 + H_i(z))| dz \leq C$, where $H_i(z) > -1, C$ is a positive constant ($i = 1, 2, 3, 4$).

Theorem 6. *For any initial value $(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)) \in \mathbb{R}_+^{4n}$, if the assumptions (A₁) and (A₂) hold, there is a unique solution $(S_1(t), I_1(t), Q_1(t), R_1(t), \dots, S_n(t), I_n(t), Q_n(t), R_n(t)) \in \mathbb{R}_+^{4n}$ for stochastic model (2) on $t \geq 0$.*

Proof. By the assumption (A₁), it can be seen that the coefficients of the diffusion term and the coefficients of the drift term satisfy the local Lipschitz condition. For any initial value $(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)) \in \mathbb{R}_+^{4n}$, stochastic model (2) has a unique local solution $(S_1(t), I_1(t), Q_1(t), R_1(t), \dots, S_n(t), I_n(t), Q_n(t), R_n(t)) \in \mathbb{R}_+^{4n}$ on $t \in [0, \tau_e)$, where τ_e is the explosion time [31]. In order to prove this solution is global, we only need to manifest that $\tau_e = \infty$, a.s. Let $d_0 > 0$ be arbitrarily large such that initial values lying with the interval $[1/d_0, d_0]$, for each $d \geq d_0$, define the stopping time as follows:

$$\tau_d = \inf\{t \in [0, \tau_e) : S_k(t) \notin (1/d, d) \text{ or } I_k(t) \notin (1/d, d) \\ \text{or } Q_k(t) \notin (1/d, d) \text{ or } R_k(t) \notin (1/d, d), k = 1, 2, \dots, n\}.$$

Let $\inf \emptyset = \infty$, and we can see that τ_d is increasing as $d \rightarrow \infty$. Define $\tau_\infty = \lim_{d \rightarrow \infty} \tau_d$, then $\tau_\infty \leq \tau_e$, a.s. Therefore, we only need to demonstrate that $\tau_\infty = \infty$, a.s. If the statement is false, there is a constant $T > 0$ and $\varepsilon \in (0, 1)$ such that $P(\tau_\infty \leq T) > \varepsilon$, so there exists an integer $d_1 \geq d_0$ such that $P(\tau_d \leq T) \geq \varepsilon$ for all $d \geq d_1$.

Define C^2 -function $V : \mathbb{R}_+^{4n} \rightarrow \mathbb{R}_+$ by

$$V(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) = \sum_{k=1}^n [(S_k - \ln S_k) + (I_k - \ln I_k) \\ + (Q_k - \ln Q_k) + (R_k - \ln R_k)].$$

Using Itô formula with jump, where L is the differential operator, we obtain

$$dV = LVdt + \sigma_1 \sum_{k=1}^n [(S_k - 1) + (I_k - 1) + (Q_k - 1) + (R_k - 1)]dB_1(t) \\ + \sigma_2 \sum_{k=1}^n [(I_k - 1) + (Q_k - 1)]dB_2(t) + \int_{\mathbb{Z}} [H_1(z)S_k(t-) - \log(1 + H_1(z)) \\ + H_2(z)I_k(t-) - \log(1 + H_2(z)) + H_3(z)Q_k(t-) - \log(1 + H_3(z)) \\ + H_4(z)R_k(t-) - \log(1 + H_4(z))] \tilde{N}(dt, dz),$$

where

$$\begin{aligned}
 LV &= \sum_{k=1}^n \left\{ \left(1 - \frac{1}{S_k}\right) \left(\Lambda - \beta k S_k \frac{\Theta}{1 + b\Theta^2} - \alpha S_k - \mu S_k \right) \right. \\
 &\quad \left. + \frac{\sigma_1^2}{2} + \int_z [H_1(z) - \log(1 + H_1(z))] \pi(dz) \right\} \\
 &\quad + \sum_{k=1}^n \left\{ \left(1 - \frac{1}{I_k}\right) \left(\beta k S_k \frac{\Theta}{1 + b\Theta^2} - (\nu + \gamma + \mu + \lambda) I_k \right) + \frac{\sigma_1^2 + \sigma_2^2}{2} \right. \\
 &\quad \left. + \int_z [H_2(z) - \log(1 + H_2(z))] \pi(dz) \right\} \\
 &\quad + \sum_{k=1}^n \left\{ \left(1 - \frac{1}{Q_k}\right) (\nu I_k - \varepsilon Q_k - \mu Q_k - \lambda Q_k) + \frac{\sigma_1^2 + \sigma_2^2}{2} \right. \\
 &\quad \left. + \int_z [H_3(z) - \log(1 + H_3(z))] \pi(dz) \right\} \\
 &\quad + \sum_{k=1}^n \left\{ \left(1 - \frac{1}{R_k}\right) (\alpha S_k + \gamma I_k + \varepsilon Q_k - \mu R_k) + \frac{\sigma_1^2}{2} \right. \\
 &\quad \left. + \int_z [H_4(z) - \log(1 + H_4(z))] \pi(dz) \right\} \\
 &\leq \sum_{k=1}^n \left[\beta k \frac{\Theta}{1 + b\Theta^2} + \Lambda + \alpha + 4\mu + \varepsilon + \nu + \gamma + 2\lambda + 2\sigma_1^2 + \sigma_2^2 + \int_z [H_1(z) \right. \\
 &\quad - \log(1 + H_1(z)) + H_2(z) - \log(1 + H_2(z)) + H_3(z) - \log(1 + H_3(z)) \\
 &\quad \left. + H_4(z) - \log(1 + H_4(z))] \pi(dz) \right].
 \end{aligned}$$

Note that

$$\sum_{k=1}^n \beta k \frac{\Theta}{1 + b\Theta^2} \leq \sum_{k=1}^n \beta k \frac{1}{\langle k \rangle} \sum_{k=1}^n k P(k) I(k) \leq \beta \frac{\langle k^2 \rangle}{\langle k \rangle} \frac{\Lambda}{\mu}.$$

Therefore

$$\begin{aligned}
 LV &\leq \sum_{k=1}^n \left[\beta \frac{\langle k^2 \rangle}{\langle k \rangle} \frac{\Lambda}{\mu} + \Lambda + 4\mu + \alpha + \varepsilon + \nu + \gamma + 2\lambda + 2\sigma_1^2 + \sigma_2^2 \right. \\
 &\quad \left. + \int_z [H_1(z) - \log(1 + H_1(z)) + H_2(z) - \log(1 + H_2(z)) + H_3(z) \right. \\
 &\quad \left. - \log(1 + H_3(z)) + H_4(z) - \log(1 + H_4(z))] \pi(dz) \right] =: K.
 \end{aligned}$$

So we get

$$\begin{aligned}
 dV \leq & K + \sigma_1 \sum_{k=1}^n [(S_k - 1) + (I_k - 1) + (Q_k - 1) + (R_k - 1)] dB_1(t) \\
 & + \sigma_2 \sum_{k=1}^n [(I_k - 1) + (Q_k - 1)] dB_2(t) + \int_z [H_1(z) S_k(t-) - \log(1 + H_1(z)) \\
 & + H_2(z) I_k(t-) - \log(1 + H_2(z)) + H_3(z) Q_k(t-) - \log(1 + H_3(z)) \\
 & + H_4(z) R_k(t-) - \log(1 + H_4(z))] \tilde{N}(dt, dz). \quad (12)
 \end{aligned}$$

Integrating both sides of Eq. (12) and taking expectation yield

$$\begin{aligned}
 & E[V(S_1(\tau_d \wedge T), I_1(\tau_d \wedge T), Q_1(\tau_d \wedge T), R_1(\tau_d \wedge T), \dots, S_n(\tau_d \wedge T), \\
 & I_n(\tau_d \wedge T), Q_n(\tau_d \wedge T), R_n(\tau_d \wedge T))] \\
 & = V[S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)] \\
 & + E\left(\int_0^{\tau_d \wedge T} LV dt\right) \\
 & \leq V[S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)] + KT. \quad (13)
 \end{aligned}$$

Let $\Omega_d = \{\tau_d \leq T\}$ for $d \geq d_1$, then $P(\Omega_d) \geq \varepsilon$. For any $\varphi \in \Omega_d$, there exists at least one of $S_k(\tau_d, \varphi)$, $I_k(\tau_d, \varphi)$, $Q_k(\tau_d, \varphi)$ and $R_k(\tau_d, \varphi)$ equaling d or d^{-1} . Consequently, we get

$$\begin{aligned}
 & V(S_1(\tau_d), I_1(\tau_d), Q_1(\tau_d), R_1(\tau_d), \dots, S_n(\tau_d), I_n(\tau_d), Q_n(\tau_d), R_n(\tau_d)) \\
 & \geq (d - 1 - \ln d) \wedge \left(\frac{1}{d} - 1 + \ln d\right). \quad (14)
 \end{aligned}$$

We can easily derive from (13) and (14) that

$$\begin{aligned}
 & V[S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)] + KT \\
 & \geq E[V(S_1(\tau_d \wedge T), I_1(\tau_d \wedge T), Q_1(\tau_d \wedge T), R_1(\tau_d \wedge T), \dots, S_n(\tau_d \wedge T), \\
 & I_n(\tau_d \wedge T), Q_n(\tau_d \wedge T), R_n(\tau_d \wedge T))] \\
 & = E[1_{\Omega_d(\varphi)} S_1(\tau_d, \varphi), I_1(\tau_d, \varphi), Q_1(\tau_d, \varphi), R_1(\tau_d, \varphi), \dots, S_n(\tau_d, \varphi), \\
 & I_n(\tau_d, \varphi), Q_n(\tau_d, \varphi), R_n(\tau_d, \varphi)] \geq \varepsilon \left[(d - 1 - \ln d) \wedge \left(\frac{1}{d} - 1 + \ln d\right) \right], \quad (15)
 \end{aligned}$$

where $1_{\Omega_d(\varphi)}$ is the indicator function of Ω_d . When $d \rightarrow \infty$, we have

$$\infty > V[S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)] + KT \geq \infty, \quad (16)$$

which leads a contradiction. So we get $\tau_\infty = \infty$, a.s. $\square$

3.2. Asymptotic behavior around the disease-free equilibrium

According to Theorem 3, for the deterministic model (1), when $R_0 < 1$, there is a disease-free equilibrium E^0 , which is globally asymptotically stable. However, E^0 is not the disease-free equilibrium of stochastic model (2) with Lévy jumps. Next, we investigate the asymptotic behavior of the solution of stochastic model (2) near E^0 with the impact of Lévy jumps.

Theorem 7. For $R_0 < 1, \omega_i > 0 (i = 1, 2, 3, 4)$, the solution $(S_1(t), I_1(t), Q_1(t), R_1(t), \dots, S_n(t), I_n(t), Q_n(t), R_n(t))$ of stochastic model (2) with initial value $(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0))$ has the following property:

$$\lim_{t \rightarrow \infty} \sup \frac{1}{t} E \int_0^t \sum_{k=1}^n \xi(k) \left[\left(S_k(r) - \frac{\Lambda}{\mu + \alpha} \right)^2 + I_k^2(r) + Q_k^2(r) + \left(R_k(r) - \frac{\alpha \Lambda}{\mu(\mu + \alpha)} \right)^2 \right] dr < \frac{K}{M},$$

where

$$\omega_1 = \mu + \frac{\alpha}{2} - \sigma_1^2 - 2 \int_z H_1^2(z) \pi(dz),$$

$$\omega_2 = \frac{\nu}{2} + \frac{\gamma}{2} + \mu + \lambda - \frac{1}{2} \sigma_1^2 - \frac{1}{2} \sigma_2^2 - \int_z H_2^2(z) \pi(dz),$$

$$\omega_3 = \frac{\varepsilon}{2} + \mu + \lambda - \frac{\nu}{2} - \frac{1}{2} \sigma_1^2 - \frac{1}{2} \sigma_2^2 - \frac{1}{2} \int_z H_3^2(z) \pi(dz),$$

$$\omega_4 = \mu - \frac{\alpha}{2} - \frac{\gamma}{2} - \frac{\varepsilon}{2} - \sigma_1^2 - \int_z H_4^2(z) \pi(dz),$$

$$K = \sum_{k=1}^n \xi(k) \left[\frac{\Lambda^2}{(\mu + \alpha)^2} (\sigma_1^2 + \int_z H_1^2(z) \pi(dz)) + \frac{\alpha^2 \Lambda^2}{\mu^2 (\mu + \alpha)^2} (\sigma_1^2 + \int_z H_4^2(z) \pi(dz)) \right],$$

$$M = \min \{ \omega_1, \omega_2, \omega_3, \omega_4 \},$$

$$\xi(k) = \frac{k^2 P(k)}{\langle k \rangle}.$$

Proof. Let $x_k = S_k - \frac{\Lambda}{\mu + \alpha}$, $y_k = I_k$, $z_k = Q_k$, $l_k = R_k - \frac{\alpha\Lambda}{\mu(\mu + \alpha)}$. Then the stochastic model (2) can be transformed into

$$\left\{ \begin{aligned} dx_k &= \left[- \left(\beta k \frac{\Theta}{1 + b\Theta^2} + \mu + \alpha \right) x_k - \beta k \frac{\Lambda}{\mu + \alpha} \frac{\Theta}{1 + b\Theta^2} \right] dt \\ &\quad + \sigma_1 \left(x_k + \frac{\Lambda}{\mu + \alpha} \right) dB_1(t) + \int_z H_1(z) \left(x_k(t-) + \frac{\Lambda}{\mu + \alpha} \right) \tilde{N}(dt, dz), \\ dy_k &= \left[\beta k \frac{\Theta}{1 + b\Theta^2} x_k + \beta k \frac{\Lambda}{\mu + \alpha} \frac{\Theta}{1 + b\Theta^2} - (\nu + \gamma + \mu + \lambda) y_k \right] dt + \sigma_1 y_k dB_1(t) \\ &\quad + \sigma_2 y_k dB_2(t) + \int_z H_2(z) y_k(t-) \tilde{N}(dt, dz), \\ dz_k &= [\nu y_k - \varepsilon z_k - \mu z_k - \lambda z_k] dt + \sigma_1 z_k dB_1(t) + \sigma_2 z_k dB_2(t) \\ &\quad + \int_z H_3(z) z_k(t-) \tilde{N}(dt, dz), \\ dl_k &= [\alpha x_k + \gamma y_k + \varepsilon z_k - \mu l_k] dt + \sigma_1 \left(l_k + \frac{\alpha\Lambda}{\mu(\mu + \alpha)} \right) dB_1(t) \\ &\quad + \int_z H_4(z) \left(l_k(t-) + \frac{\alpha\Lambda}{\mu(\mu + \alpha)} \right) \tilde{N}(dt, dz). \end{aligned} \right.$$

Define C^2 -function $U : \mathbb{R}_+^{4n} \rightarrow \mathbb{R}_+$ by

$$\begin{aligned} U(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \\ &= \frac{1}{2} \sum_{k=1}^n \xi(k) (x_k + y_k)^2 + p y_k + \frac{1}{2} \sum_{k=1}^n \xi(k) z_k^2 + \frac{1}{2} \sum_{k=1}^n \xi(k) l_k^2 \\ &=: U_1 + p U_2 + U_3 + U_4, \end{aligned}$$

where $p = \frac{\nu + \gamma + \lambda + \alpha + 2\mu}{\beta}$.

Applying Itô formula with jump for $U(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n)$, we can obtain

$$\begin{aligned} dU(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \\ &= LU_1 dt + \sum_{k=1}^n \xi(k) (x_k + y_k) \left[\sigma_1 \left(x_k + \frac{\Lambda}{\mu + \alpha} \right) dB_1(t) \right. \\ &\quad \left. + \sigma_1 y_k dB_1(t) + \sigma_2 y_k dB_2(t) \right] + \sum_{k=1}^n \xi(k) \int_z \\ &\quad \left\{ \frac{1}{2} \left[H_1(z) \left(x_k(t-) + \frac{\Lambda}{\mu + \alpha} \right) + H_2(z) y_k(t-) \right]^2 \right. \end{aligned}$$

$$\begin{aligned}
& + (x_k(t-) + y_k(t-)) \left[H_1(z) \left(x_k(t-) + \frac{\Lambda}{\mu + \alpha} \right) + H_2(z) y_k(t-) \right] \Big\} \tilde{N}(dt, dz) \\
& + p L U_2 dt + p \sum_{k=1}^n \xi(k) \int_z H_2(z) y_k(t-) \tilde{N}(dt, dz) + L U_3 dt \\
& + \sum_{k=1}^n \xi(k) [\sigma_1 z_k^2 dB_1(t) + \sigma_2 z_k^2 dB_2(t)] \\
& + \sum_{k=1}^n \xi(k) \int_z \left(\frac{1}{2} H_3^2(z) z_k^2(t-) + H_3(z) z_k^2(t-) \right) \tilde{N}(dt, dz) \\
& + L U_4 dt + \sum_{k=1}^n \xi(k) l_k \sigma_1 \left(l_k + \frac{\alpha \Lambda}{\mu(\mu + \alpha)} \right) dB_1(t) \\
& + \sum_{k=1}^n \xi(k) \int_z \left\{ \frac{1}{2} \left[H_4(z) \left(l_k(t-) + \frac{\alpha \Lambda}{\mu(\mu + \alpha)} \right) \right]^2 \right. \\
& \left. + l_k \left[l_k + H_4(z) \left(l_k(t-) + \frac{\alpha \Lambda}{\mu(\mu + \alpha)} \right) \right] \right\} \tilde{N}(dt, dz),
\end{aligned}$$

where

$$\begin{aligned}
L U_1 &= \sum_{k=1}^n \xi(k) (x_k + y_k) [-(\mu + \alpha) x_k - (\nu + \gamma + \mu + \lambda) y_k] \\
&+ \frac{1}{2} \sum_{k=1}^n \xi(k) \left[\sigma_1^2 \left(x_k + \frac{\Lambda}{\mu + \alpha} \right)^2 + \sigma_1^2 y_k^2 + \sigma_2^2 y_k^2 \right] \\
&+ \frac{1}{2} \sum_{k=1}^n \xi(k) \int_z \left[H_1(z) \left(x_k + \frac{\Lambda}{\mu + \alpha} \right) + H_2(z) y_k \right]^2 \pi(dz) \\
&= \sum_{k=1}^n \xi(k) [-(\mu + \alpha) x_k^2 - (\nu + \gamma + \mu + \lambda) y_k^2 - (\nu + \gamma + \lambda + \alpha + 2\mu) x_k y_k] \\
&+ \sum_{k=1}^n \xi(k) \left[\frac{1}{2} \sigma_1^2 x_k^2 + \frac{1}{2} \sigma_1^2 \left(\frac{\Lambda}{\mu + \alpha} \right)^2 + \sigma_1^2 x_k \frac{\Lambda}{\mu + \alpha} + \frac{1}{2} \sigma_1^2 y_k^2 + \frac{1}{2} \sigma_2^2 y_k^2 \right] \\
&+ \sum_{k=1}^n \xi(k) \left[\frac{1}{2} \int_z H_1(z) \left(x_k + \frac{\Lambda}{\mu + \alpha} \right) + H_2(z) y_k \right]^2 \pi(dz) \\
&\leq - \left(\mu + \alpha - \sigma_1^2 - 2 \int_z H_1^2(z) \pi(dz) \right) \sum_{k=1}^n \xi(k) x_k^2
\end{aligned}$$

$$\begin{aligned}
 & - \left(\nu + \gamma + \mu + \lambda - \frac{1}{2}\sigma_1^2 - \frac{1}{2}\sigma_2^2 - \int_z H_2^2(z)\pi(dz) \right) \sum_{k=1}^n \xi(k)y_k^2 \\
 & - (\nu + \gamma + \lambda + \alpha + 2\mu) \sum_{k=1}^n \xi(k)x_k y_k \\
 & + \left(\sigma_1^2 \frac{\Lambda^2}{(\mu + \alpha)^2} + \int_z H_1^2(z) \frac{\Lambda^2}{(\mu + \alpha)^2} \pi(dz) \right), \\
 LU_2 &= \beta k \frac{\Theta}{1 + b\Theta^2} x_k + \beta k \frac{\Lambda}{\mu + \alpha} \frac{\Theta}{1 + b\Theta^2} - (\nu + \gamma + \mu + \lambda) y_k \\
 & \leq \beta k \frac{1}{\langle k \rangle} \sum_{k=1}^n k P(k) x_k y_k + \beta k \frac{\Lambda}{\mu + \alpha} \frac{1}{\langle k \rangle} \sum_{k=1}^n k P(k) y_k - (\nu + \gamma + \mu + \lambda) y_k \\
 & = \beta \sum_{k=1}^n \frac{k^2 P(k)}{\langle k \rangle} x_k y_k + (\nu + \gamma + \mu + \lambda)(R_0 - 1) y_k \leq \beta \sum_{k=1}^n \frac{k^2 P(k)}{\langle k \rangle} x_k y_k, \\
 LU_3 &= \sum_{k=1}^n \xi(k) \left[z_k (\nu y_k - \varepsilon z_k - \mu z_k - \lambda z_k) + \frac{1}{2} (\sigma_1^2 z_k^2 + \sigma_2^2 z_k^2) + \frac{1}{2} \int_z H_3^2(z) z_k^2 \pi(dz) \right] \\
 & \leq \sum_{k=1}^n \xi(k) \left[\frac{\nu}{2} y_k^2 - \left(\varepsilon + \mu + \lambda - \frac{\nu}{2} \right) z_k^2 + \frac{1}{2} (\sigma_1^2 z_k^2 + \sigma_2^2 z_k^2) + \frac{1}{2} \int_z H_3^2(z) z_k^2 \pi(dz) \right] \\
 & = \sum_{k=1}^n \xi(k) \left[\frac{\nu}{2} y_k^2 - \left(\varepsilon + \mu + \lambda - \frac{\nu}{2} - \frac{1}{2}\sigma_1^2 - \frac{1}{2}\sigma_2^2 - \frac{1}{2} \int_z H_3^2(z) \pi(dz) \right) z_k^2 \right], \\
 LU_4 &= \sum_{k=1}^n \xi(k) \left[l_k (\alpha x_k + \gamma y_k + \varepsilon z_k - \mu l_k) + \frac{1}{2} \sigma_1^2 \left(l_k + \frac{\alpha \Lambda}{\mu(\mu + \alpha)} \right)^2 \right. \\
 & \quad \left. + \frac{1}{2} \int_z H_4^2(z) \left(l_k + \frac{\alpha \Lambda}{\mu(\mu + \alpha)} \right)^2 \pi(dz) \right] \\
 & \leq \sum_{k=1}^n \xi(k) \left[\frac{\alpha}{2} x_k^2 + \frac{\gamma}{2} y_k^2 + \frac{\varepsilon}{2} z_k^2 - \left(\mu - \frac{\alpha}{2} - \frac{\gamma}{2} - \frac{\varepsilon}{2} \right) l_k^2 \right. \\
 & \quad \left. + \sigma_1^2 l_k^2 + \sigma_1^2 \frac{\alpha^2 \Lambda^2}{\mu^2 (\mu + \alpha)^2} + \int_z \left(H_4^2(z) l_k^2 + H_4^2 \frac{\alpha^2 \Lambda^2}{\mu^2 (\mu + \alpha)^2} \right) \pi(dz) \right] \\
 & = \sum_{k=1}^n \xi(k) \left[\frac{\alpha}{2} x_k^2 + \frac{\gamma}{2} y_k^2 + \frac{\varepsilon}{2} z_k^2 - \left(\mu - \frac{\alpha}{2} - \frac{\gamma}{2} - \frac{\varepsilon}{2} - \sigma_1^2 - \int_z H_4^2(z) \pi(dz) \right) l_k^2 \right. \\
 & \quad \left. + \sigma_1^2 \frac{\alpha^2 \Lambda^2}{\mu^2 (\mu + \alpha)^2} + \int_z H_4^2 \frac{\alpha^2 \Lambda^2}{\mu^2 (\mu + \alpha)^2} \pi(dz) \right],
 \end{aligned}$$

therefore

$$\begin{aligned}
& LU(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \\
&= LU_1 + pLU_2 + LU_3 + LU_4 \\
&\leq \left(\mu + \frac{\alpha}{2} - \sigma_1^2 - 2 \int_z H_1^2(z) \pi(dz) \right) \sum_{k=1}^n \xi(k) x_k^2 \\
&\quad - \left(\frac{\nu}{2} + \frac{\gamma}{2} + \mu + \lambda - \frac{1}{2} \sigma_1^2 - \frac{1}{2} \sigma_2^2 - \int_z H_2^2(z) \pi(dz) \right) \sum_{k=1}^n \xi(k) y_k^2 \\
&\quad - \left(\frac{\varepsilon}{2} + \mu + \lambda - \frac{\nu}{2} - \frac{1}{2} \sigma_1^2 - \frac{1}{2} \sigma_2^2 - \frac{1}{2} \int_z H_3^2(z) \pi(dz) \right) \sum_{k=1}^n \xi(k) z_k^2 \\
&\quad - \left(\mu - \frac{\alpha}{2} - \frac{\gamma}{2} - \frac{\varepsilon}{2} - \sigma_1^2 - \int_z H_4^2(z) \pi(dz) \right) \sum_{k=1}^n \xi(k) l_k^2 \\
&\quad + \sum_{k=1}^n \xi(k) \left[\frac{\Lambda^2}{(\mu + \alpha)^2} (\sigma_1^2 + \int_z H_1^2(z) \pi(dz)) \right. \\
&\quad \left. + \frac{\alpha^2 \Lambda^2}{\mu^2 (\mu + \alpha)^2} \left(\sigma_1^2 + \int_z H_4^2(z) \pi(dz) \right) \right].
\end{aligned}$$

Denote $\sum_{k=1}^n \xi(k) \left[\frac{\Lambda^2}{(\mu + \alpha)^2} (\sigma_1^2 + \int_z H_1^2(z) \pi(dz)) + \frac{\alpha^2 \Lambda^2}{\mu^2 (\mu + \alpha)^2} (\sigma_1^2 + \int_z H_4^2(z) \pi(dz)) \right] := K$, and we obtain

$$\begin{aligned}
& LU(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \\
&\leq \sum_{k=1}^n \xi(k) (-\omega_1 x_k^2 - \omega_2 y_k^2 - \omega_3 z_k^2 - \omega_4 l_k^2) + K. \tag{17}
\end{aligned}$$

We integrate both sides of Eq. (17) and take the expectation as follows:

$$\begin{aligned}
0 &\leq E[U(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n)] \\
&\leq U(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)) \\
&\quad + E \int_0^t \left[\sum_{k=1}^n \xi(k) (-\omega_1 x_k^2(r) - \omega_2 y_k^2(r) - \omega_3 z_k^2(r) - \omega_4 l_k^2(r)) + K \right] dr,
\end{aligned}$$

that is

$$\begin{aligned}
& E \int_0^t \left[\sum_{k=1}^n \xi(k) (\omega_1 x_k^2(r) + \omega_2 y_k^2(r) + \omega_3 z_k^2(r) + \omega_4 l_k^2(r)) \right] dr \\
&< U(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)) + Kt.
\end{aligned}$$

It follows that

$$\lim_{t \rightarrow \infty} \sup \frac{1}{t} E \int_0^t \left[\sum_{k=1}^n \xi(k) (\omega_1 x_k^2(r) + \omega_2 y_k^2(r) + \omega_3 z_k^2(r) + \omega_4 l_k^2(r)) \right] dr < K.$$

Let $M = \min\{\omega_1, \omega_2, \omega_3, \omega_4\}$, and one can easily derive that

$$\lim_{t \rightarrow \infty} \sup \frac{1}{t} E \int_0^t \left[\sum_{k=1}^n \xi(k) (x_k^2(r) + y_k^2(r) + z_k^2(r) + l_k^2(r)) \right] dr < \frac{K}{M},$$

that is

$$\begin{aligned} \lim_{t \rightarrow \infty} \sup \frac{1}{t} E \int_0^t \left[\sum_{k=1}^n \xi(k) \left(\left(S_k(r) - \frac{\Lambda}{\mu + \alpha} \right)^2 + I_k^2(r) + Q_k^2(r) \right. \right. \\ \left. \left. + \left(R_k(r) - \frac{\alpha \Lambda}{\mu(\mu + \alpha)} \right)^2 \right) \right] dr < \frac{K}{M}. \end{aligned} \quad \square$$

Remark 1. The basic reproduction number $R_0 = \frac{\langle k^2 \rangle}{\langle k \rangle} \frac{\beta \Lambda}{(\mu + \alpha)(\nu + \gamma + \lambda + \mu)}$ means that epidemic threshold depends on the level of heterogeneity of the network ($\langle k^2 \rangle / \langle k \rangle$), transmission rate (β), recruitment rate (Λ), infected individuals being cured rate (γ), quarantined individuals being cured rate (ε), vaccinate rate (α), natural death rate (μ), disease-related death rate (λ) and quarantine rate (ν). Then if the heterogeneity of the network, transmission rate and recruitment rate decrease, or natural death rate, vaccinate rate, quarantine rate, infected individuals being cured rate and disease-related death rate increase, the corresponding basic reproduction number R_0 decreases. Especially, when $R_0 < 1$, the solution of the stochastic model (2) fluctuates around the disease-free equilibrium.

Remark 2. The intensity of the fluctuation is related to $\sigma_i (i = 1, 2)$, $H_j(z)$ ($j = 1, 2, 3, 4$), the smaller values of σ_i and $H_j(z)$, and the weaker fluctuation of trivial solution of stochastic model (2). That is, if σ_i and $H_j(z)$ decrease, the solution of stochastic model (2) will approach to the disease-free equilibrium E^0 , which leads to the eradication of the disease.

3.3. Asymptotic behavior around the endemic equilibrium

From Theorem 5, there is an endemic equilibrium E^* for the deterministic model (1) when $R_0 > 1$, which is globally asymptotically stable. However, E^* is not the endemic equilibrium of stochastic model (2) with Lévy jumps. Then we study the asymptotic behavior of the solution of stochastic model (2) near E^* with the impact of Lévy jumps.

Theorem 8. When $R_0 > 1, \eta_i > 0$ ($i = 1, 2, 3, 4$), the solution $(S_1(t), I_1(t), Q_1(t), R_1(t), \dots, S_n(t), I_n(t), Q_n(t), R_n(t))$ of stochastic model (2) with initial value

$(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0))$ has the following property:

$$\lim_{t \rightarrow \infty} \sup \frac{1}{t} E \int_0^t \sum_{k=1}^n \psi(k) [(S_k(r) - S_k^*)^2 + (I_k(r) - I_k^*)^2 + (Q_k(r) - Q_k^*)^2 + (R_k(r) - R_k^*)^2] dr < \frac{L}{N},$$

where

$$\begin{aligned} L = & \sum_{k=1}^n \psi(k) [\sigma_1^2 (S_k^*)^2 + (\sigma_1^2 + \sigma_2^2) (I_k^*)^2 + (\sigma_1^2 + \sigma_2^2) (Q_k^*)^2 + \sigma_1^2 (R_k^*)^2 \\ & + p^* I_k^* \int_z (H_2(z) - \ln(1 + H_2(z))) \pi(dz) + \frac{1}{2} p^* I_k^* (\sigma_1^2 + \sigma_2^2) \\ & + \int_z H_1^2(z) (S_k^*)^2 \pi(dz) + \int_z H_2^2(z) (I_k^*)^2 \pi(dz) \\ & + \int_z H_3^2(z) (Q_k^*)^2 \pi(dz) + \int_z H_4^2(z) (R_k^*)^2 \pi(dz)], \end{aligned}$$

$$\psi(k) = \frac{kP(k)}{\langle k \rangle},$$

$$N = \min\{\eta_1, \eta_2, \eta_3, \eta_4\},$$

$$p^* = \frac{(\nu + \gamma + \lambda + \alpha + 2\mu)(1 + b(\Theta^*)^2)\langle k \rangle}{\beta \langle k^2 \rangle},$$

$$\eta_1 = \mu + \frac{\alpha}{2} - \sigma_1^2 - \int_z H_1^2(z) \pi(dz),$$

$$\eta_2 = \frac{\nu}{2} + \frac{\gamma}{2} + \mu + \lambda - \sigma_1^2 - \sigma_2^2 - \int_z H_2^2(z) \pi(dz),$$

$$\eta_3 = \frac{\varepsilon}{2} + \mu + \lambda - \frac{\nu}{2} - \sigma_1^2 - \sigma_2^2 - \int_z H_3^2(z) \pi(dz),$$

$$\eta_4 = \mu - \frac{\alpha}{2} - \frac{\gamma}{2} - \frac{\varepsilon}{2} - \sigma_1^2 - \int_z H_4^2(z) \pi(dz).$$

Proof. If $R_0 > 1$, the deterministic model corresponding to stochastic model (2) has a unique endemic equilibrium, so we have

$$\begin{cases} \alpha S_k^* + \mu S_k^* + \nu I_k^* + \gamma I_k^* + \mu I_k^* + \lambda I_k^* = \Lambda, \\ \nu I_k^* - \varepsilon Q_k^* - \mu Q_k^* - \lambda Q_k^* = 0, \\ \alpha S_k^* + \gamma I_k^* + \varepsilon Q_k^* - \mu R_k^* = 0. \end{cases}$$

Define C^2 -function $Y : \mathbb{R}_+^{4n} \rightarrow \mathbb{R}_+$ by

$$\begin{aligned} Y(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \\ &= \frac{1}{2} \sum_{k=1}^n \psi(k) (S_k - S_k^* + I_k - I_k^*)^2 + p^* \left(\Theta - \Theta^* - \Theta^* \ln \frac{\Theta}{\Theta^*} \right) \\ &\quad + \frac{1}{2} \sum_{k=1}^n \psi(k) (Q_k - Q_k^*)^2 + \frac{1}{2} \sum_{k=1}^n \psi(k) (R_k - R_k^*)^2 \\ &=: Y_1 + p^* Y_2 + Y_3 + Y_4, \end{aligned}$$

where $p^* = \frac{(\nu + \gamma + \lambda + \alpha + 2\mu)(1 + b(\Theta^*)^2)\langle k \rangle}{\beta \langle k^2 \rangle}$.

Applying Itô formula with jump for $Y(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n)$, we can obtain

$$\begin{aligned} dY(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \\ &= LY_1 dt + \sum_{k=1}^n \psi(k) (S_k - S_k^* + I_k - I_k^*) [\sigma_1 S_k dB_1(t) \\ &\quad + \sigma_1 I_k dB_1(t) + \sigma_2 I_k dB_2(t)] \\ &\quad + \sum_{k=1}^n \psi(k) \int_z \left[\frac{1}{2} (H_1(z) S_k(t-) + H_2(z) I_k(t-))^2 + (H_1(z) S_k(t-) \right. \\ &\quad \left. + H_2(z) I_k(t-)) (S_k(t-) - S_k^* + I_k(t-) - I_k^*) \right] \tilde{N}(dt, dz) \\ &\quad + LY_2 dt + \sum_{k=1}^n \psi(k) \left(1 - \frac{\Theta^*}{\Theta} \right) [\sigma_1 I_k dB_1(t) + \sigma_2 I_k dB_2(t)] \\ &\quad + \sum_{k=1}^n \psi(k) \int_z [H_2(z) I_k(t-) - I_k^* \ln(1 + H_2(z))] \tilde{N}(dt, dz) \\ &\quad + LY_3 dt + \sum_{k=1}^n \psi(k) (Q_k - Q_k^*) [\sigma_1 Q_k dB_1(t) + \sigma_2 Q_k dB_2(t)] \\ &\quad + \sum_{k=1}^n \psi(k) \int_z \left[\frac{1}{2} H_3^2(z) Q_k^2(t-) + (Q_k(t-) - Q_k^*) H_3(z) Q_k(t-) \right] \tilde{N}(dt, dz) \\ &\quad + LY_4 dt + \sum_{k=1}^n \psi(k) (R_k - R_k^*) \sigma_1 R_k dB_1(t) \\ &\quad + \sum_{k=1}^n \psi(k) \int_z \left[\frac{1}{2} H_4^2(z) R_k^2(t-) + (R_k(t-) - R_k^*) H_4(z) R_k(t-) \right] \tilde{N}(dt, dz), \end{aligned}$$

where

$$\begin{aligned}
 LY_1 &= \sum_{k=1}^n \psi(k)(S_k - S_k^* + I_k - I_k^*)(\Lambda - \alpha S_k - \mu S_k - \nu I_k - \gamma I_k - \mu I_k - \lambda I_k) \\
 &\quad + \frac{1}{2} \sum_{k=1}^n \psi(k)(\sigma_1^2 S_k^2 + \sigma_1^2 I_k^2 + \sigma_2^2 I_k^2) + \frac{1}{2} \sum_{k=1}^n \psi(k) \\
 &\quad \times \int_z (H_1(z)S_k + H_2(z)I_k)^2 \pi(dz) \\
 &\leq -(\alpha + \mu - \sigma_1^2 - \int_z H_1^2(z)\pi(dz)) \sum_{k=1}^n \psi(k)(S_k - S_k^*)^2 \\
 &\quad -(\nu + \gamma + \mu + \lambda - \sigma_1^2 - \sigma_2^2 - \int_z H_2^2(z)\pi(dz)) \sum_{k=1}^n \psi(k)(I_k - I_k^*)^2 \\
 &\quad -(\alpha + 2\mu + \nu + \gamma + \lambda)(S_k - S_k^*)(\Theta - \Theta^*) \\
 &\quad + \sum_{k=1}^n \psi(k) \left[\sigma_1^2 (S_k^*)^2 + \sigma_1^2 (I_k^*)^2 + \sigma_2^2 (I_k^*)^2 + \int_z H_1^2(z)(S_k^*)^2 \pi(dz) \right. \\
 &\quad \left. + \int_z H_2^2(z)(I_k^*)^2 \pi(dz) \right].
 \end{aligned}$$

$$\begin{aligned}
 LY_2 &= \left(1 - \frac{\Theta^*}{\Theta}\right) \sum_{k=1}^n \frac{kP(k)}{\langle k \rangle} \left[\beta k S_k \frac{\Theta}{1 + b\Theta^2} - (\nu + \gamma + \mu + \lambda) I_k \right] \\
 &\quad + \frac{1}{2} \sum_{k=1}^n \frac{\Theta^*}{\Theta^2} \left(\frac{kP(k)}{\langle k \rangle} I_k \right)^2 (\sigma_1^2 + \sigma_2^2) + \Theta^* \int_z [H_2(z) - \ln(1 + H_2(z))] \pi(dz) \\
 &\leq \sum_{k=1}^n \frac{\langle k^2 \rangle}{\langle k \rangle} \frac{\beta}{1 + b(\Theta^*)^2} (S_k - S_k^*)(\Theta - \Theta^*) + \frac{1}{2} \sum_{k=1}^n \Theta^* (\sigma_1^2 + \sigma_2^2) \\
 &\quad + \Theta^* \int_z [H_2(z) - \ln(1 + H_2(z))] \pi(dz),
 \end{aligned}$$

$$\begin{aligned}
 LY_3 &= \sum_{k=1}^n \psi(k)(Q_k - Q_k^*)(\nu I_k - \varepsilon Q_k - \mu Q_k - \lambda Q_k) \\
 &\quad + \frac{1}{2} \sum_{k=1}^n \psi(k)(\sigma_1^2 Q_k^2 + \sigma_2^2 Q_k^2) + \frac{1}{2} \sum_{k=1}^n \psi(k) \int_z H_3^2(z) Q_k^2 \pi(dz) \\
 &\leq \frac{\nu}{2} \sum_{k=1}^n \psi(k)(I_k - I_k^*)^2 - \left(\varepsilon + \mu + \lambda - \frac{\nu}{2} - \sigma_1^2 - \sigma_2^2 - \int_z H_3^2(z)\pi(dz) \right)
 \end{aligned}$$

$$\begin{aligned} & \times \sum_{k=1}^n \psi(k)(Q_k - Q_k^*)^2 + \sum_{k=1}^n \psi(k) \left[\sigma_1^2 (Q_k^*)^2 + \sigma_2^2 (Q_k^*)^2 \right. \\ & \left. + \int_z H_3^2(z) (Q_k^*)^2 \pi(dz) \right], \end{aligned}$$

$$\begin{aligned} LY_4 &= \sum_{k=1}^n \psi(k)(R_k - R_k^*)(\alpha S_k + \gamma I_k + \varepsilon Q_k - \mu R_k) + \frac{1}{2} \sum_{k=1}^n \psi(k) \sigma_1^2 R_k^2 \\ &+ \frac{1}{2} \sum_{k=1}^n \psi(k) \int_z H_4^2(z) R_k^2 \pi(dz) \\ &\leq \frac{\alpha}{2} \sum_{k=1}^n \psi(k)(S_k - S_k^*)^2 + \frac{\gamma}{2} \sum_{k=1}^n \psi(k)(I_k - I_k^*)^2 + \frac{\varepsilon}{2} \sum_{k=1}^n \psi(k)(Q_k - Q_k^*)^2 \\ &- \left(\mu - \frac{\alpha}{2} - \frac{\gamma}{2} - \frac{\varepsilon}{2} - \sigma_1^2 - \int_z H_4^2(z) R_k^2 \pi(dz) \right) \sum_{k=1}^n \psi(k)(R_k - R_k^*)^2 \\ &+ \sum_{k=1}^n \psi(k) \left[\sigma_1^2 (R_k^*)^2 + \int_z H_4^2(z) (R_k^*)^2 \pi(dz) \right]. \end{aligned}$$

Therefore, we get

$$\begin{aligned} LY &= LY_1 + p^* LY_2 + LY_3 + LY_4 \\ &\leq \left(\mu + \frac{\alpha}{2} - \sigma_1^2 - \int_z H_1^2(z) \pi(dz) \right) \sum_{k=1}^n \psi(k)(S_k - S_k^*)^2 \\ &- \left(\frac{\nu}{2} + \frac{\gamma}{2} + \mu + \lambda - \sigma_1^2 - \sigma_2^2 - \int_z H_2^2(z) \pi(dz) \right) \sum_{k=1}^n \psi(k)(I_k - I_k^*)^2 \\ &- \left(\frac{\varepsilon}{2} + \mu + \lambda - \frac{\nu}{2} - \sigma_1^2 - \sigma_2^2 - \int_z H_3^2(z) \pi(dz) \right) \sum_{k=1}^n \psi(k)(Q_k - Q_k^*)^2 \\ &- \left(\mu - \frac{\alpha}{2} - \frac{\gamma}{2} - \frac{\varepsilon}{2} - \sigma_1^2 - \int_z H_4^2(z) \pi(dz) \right) \sum_{k=1}^n \psi(k)(R_k - R_k^*)^2 + L. \end{aligned}$$

Here

$$\begin{aligned} L &:= \sum_{k=1}^n \psi(k) \left[\sigma_1^2 (S_k^*)^2 + (\sigma_1^2 + \sigma_2^2) (I_k^*)^2 + (\sigma_1^2 + \sigma_2^2) (Q_k^*)^2 + \sigma_1^2 (R_k^*)^2 + p^* I_k^* \right. \\ &\times \int_z (H_2(z) - \ln(1 + H_2(z))) \pi(dz) + \frac{1}{2} p^* I_k^* (\sigma_1^2 + \sigma_2^2) + \int_z H_1^2(z) (S_k^*)^2 \pi(dz) \\ &\left. + \int_z H_2^2(z) (I_k^*)^2 \pi(dz) + \int_z H_3^2(z) (Q_k^*)^2 \pi(dz) + \int_z H_4^2(z) (R_k^*)^2 \pi(dz) \right]. \end{aligned}$$

and we can easily obtain that

$$\begin{aligned} &LY(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n) \\ &\leq \sum_{k=1}^n \psi(k) (-\eta_1(S_k - S_k^*)^2 - \eta_2(I_k - I_k^*)^2 \\ &\quad - \eta_3(Q_k - Q_k^*)^2 - \eta_4(R_k - R_k^*)^2) + L, \end{aligned} \quad (18)$$

integrating both sides of Eq. (18) and taking the expectation as follows:

$$\begin{aligned} 0 &\leq E[Y(S_1, I_1, Q_1, R_1, \dots, S_n, I_n, Q_n, R_n)] \\ &\leq Y(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)) \\ &\quad + E \int_0^t \left[\sum_{k=1}^n \psi(k) (-\eta_1(S_k(r) - S_k^*)^2 - \eta_2(I_k(r) - I_k^*)^2 \right. \\ &\quad \left. - \eta_3(Q_k(r) - Q_k^*)^2 - \eta_4(R_k(r) - R_k^*)^2) + L \right] dr, \end{aligned}$$

that is,

$$\begin{aligned} &E \int_0^t \left[\sum_{k=1}^n \psi(k) (\eta_1(S_k(r) - S_k^*)^2 + \eta_2(I_k(r) - I_k^*)^2 + \eta_3(Q_k(r) - Q_k^*)^2 \right. \\ &\quad \left. + \eta_4(R_k(r) - R_k^*)^2) \right] dr \\ &< Y(S_1(0), I_1(0), Q_1(0), R_1(0), \dots, S_n(0), I_n(0), Q_n(0), R_n(0)) + Lt. \end{aligned}$$

It follows that

$$\begin{aligned} &\limsup_{t \rightarrow \infty} \frac{1}{t} E \int_0^t \left[\sum_{k=1}^n \psi(k) (\eta_1(S_k(r) - S_k^*)^2 + \eta_2(I_k(r) - I_k^*)^2 \right. \\ &\quad \left. + \eta_3(Q_k(r) - Q_k^*)^2 + \eta_4(R_k(r) - R_k^*)^2) \right] dr < L. \end{aligned}$$

Let $N = \min\{\eta_1, \eta_2, \eta_3, \eta_4\}$, and we can obtain that

$$\begin{aligned} &\limsup_{t \rightarrow \infty} \frac{1}{t} E \int_0^t \left[\sum_{k=1}^n \psi(k) ((S_k(r) - S_k^*)^2 + (I_k(r) - I_k^*)^2 \right. \\ &\quad \left. + (Q_k(r) - Q_k^*)^2 + (R_k(r) - R_k^*)^2) \right] dr < \frac{L}{N}. \end{aligned} \quad \square$$

Remark 3. According to $R_0 = \frac{\langle k^2 \rangle}{\langle k \rangle} \frac{\beta \Lambda}{(\mu + \alpha)(\nu + \gamma + \lambda + \mu)}$, we know that if the heterogeneity of the network ($\langle k^2 \rangle / \langle k \rangle$), transmission rate (β) and recruitment rate (Λ)

increase, or natural death rate (μ), vaccinate rate (α), quarantine rate (ν), infected individuals being cured rate (γ) and disease-related death rate (λ) increase, the corresponding basic reproduction number R_0 increase. Especially, when $R_0 > 1$, the solution of the stochastic model (2) fluctuates around the endemic equilibrium.

Remark 4. The intensity of the fluctuation is related to $\sigma_i (i = 1, 2), H_j(z)$ ($j = 1, 2, 3, 4$), the smaller values of σ_i and $H_j(z)$, and the weaker fluctuation of nontrivial solution of stochastic model (2). That is, with the decreasing of the values of σ_i and $H_j(z)$, the solution of stochastic model (2) will approach to the endemic equilibrium E^* , the disease will persist.

4. Numerical Experiments

In this part, two numerical experiments are shown to illustrate the validity of our obtained theoretical results. By using preferential attachment mechanism in [1], a scale-free network Q with the number of nodes on network is $N = 100$ and $P(k) = (r - 1)m^{r-1}k^{-r}$, ($m = 1, r = 3$), we can easily obtain that $\langle k \rangle = 3.27, \langle k^2 \rangle = 9.04$.

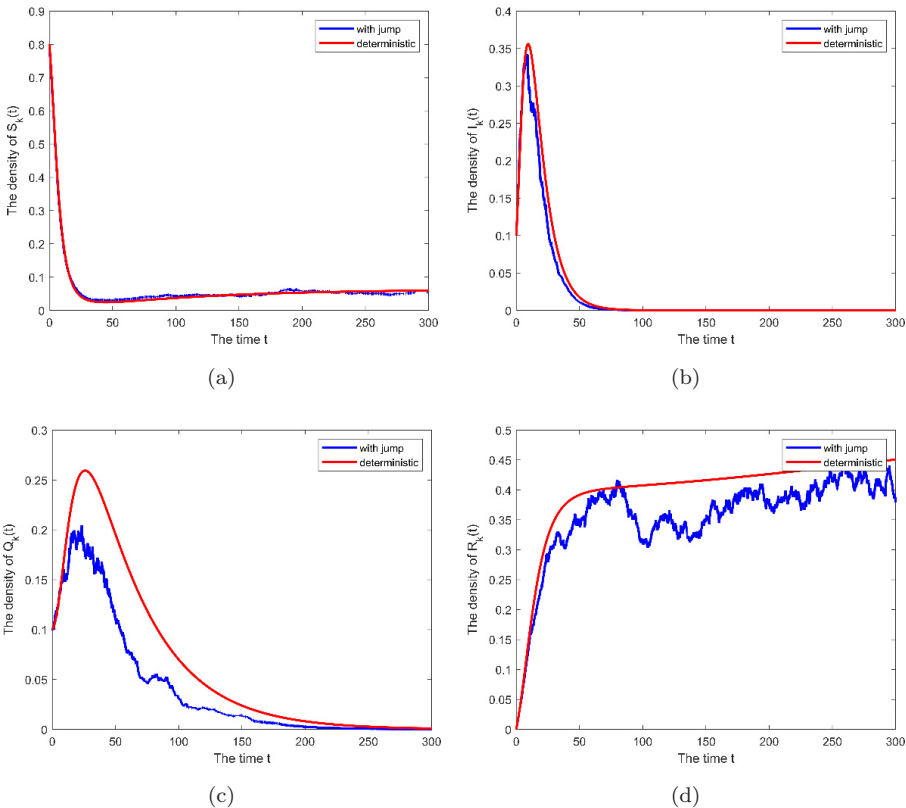


Fig. 2. When $R_0 < 1$, the trajectories of the stochastic model (2) and its corresponding deterministic model (1).

Example 1. Based on Q, the basic reproduction number $R_0 \approx 0.95 < 1$ is given with the parameters $\beta = 0.54$, $\alpha = 0.0088$, $\mu = 0.0006$, $\nu = 0.04$, $\gamma = 0.04$, $\lambda = 0.02$, $\varepsilon = 0.001$, $b = 3$, $\Lambda = 0.0006$, $\sigma_1 = \sigma_2 = 0.02$, $H_1(z) = H_2(z) = H_3(z) = H_4(z) = 0.002$. So the given parameters satisfy the conditions in Theorem 7 and deterministic model (1) has disease-free equilibrium, in Fig. 2, the red lines and blue lines of (a)–(d) represent the trajectories of the susceptible, infected, isolated and recovered individuals in the deterministic model (1) and the stochastic model (2), respectively. Obviously, the blue line fluctuates near the red line, that is, the solution of stochastic model (2) fluctuates around the disease-free equilibrium E^0 of the deterministic model (1). In particular, it can be seen from Fig. 2(b) that diseases under the influence of Lévy jumps are more likely to eradicate.

Example 2. Based on Q, the basic reproduction number $R_0 = \frac{\langle k^2 \rangle}{\langle k \rangle} \times \frac{\beta \Lambda}{(\mu + \alpha)(\nu + \gamma + \lambda + \mu)} \approx 7.15 > 1$ is given with the parameters $\beta = 0.75$, $\alpha = 0.01$, $\mu = 0.0235$, $\nu = 0.08$, $\gamma = 0.05$, $\lambda = 0.05$, $\varepsilon = 0.001$, $b = 3$, $\Lambda = 0.0235$, $\sigma_1 = \sigma_2 = 0.02$, $H_1(z) = H_2(z) = H_3(z) = H_4(z) = 0.002$. So the conditions in Theorem 8 are

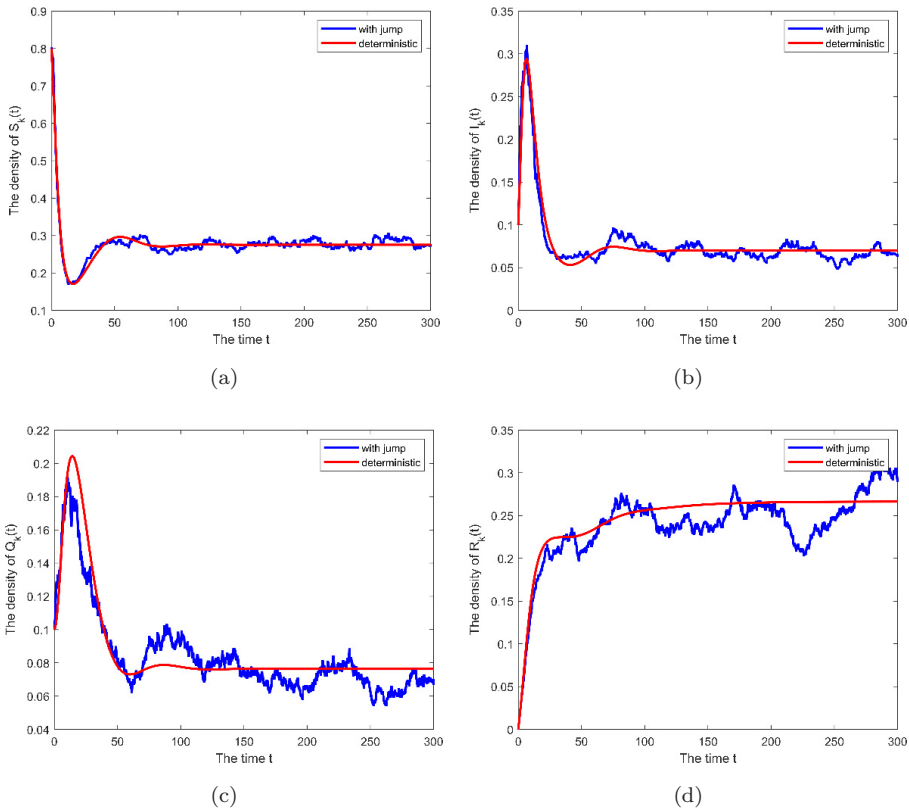


Fig. 3. When $R_0 > 1$, the trajectories of the stochastic model (2) and its corresponding deterministic model (1).

satisfied and deterministic model (1) has a unique endemic equilibrium, in Fig. 3, the red lines and blue lines of (a)–(h) represent the trajectories of the susceptible, infected, isolated and recovered individuals in the deterministic model (1) and the stochastic model (2), respectively. Obviously, the blue line fluctuates around the red line, that is, the solution of stochastic model (2) fluctuates around the endemic equilibrium E^* of the deterministic model (1).

5. Discussion

A series of achievements have been made in the research on the dynamic behavior of disease, but most studies assume that individuals are uniformly mixed. However, social networks are not uniform networks, exhibit scale-free properties. Furthermore, in the absence of effective medicine, it is necessary to take non-pharmaceutical control measures. In addition, not only non-accidental changes (temperature, humidity, etc.) can have an effect on disease dynamics, but also contingent changes (earthquakes, floods, tsunamis, etc.). To comprehensively explore the effects of the above factors on disease dynamics, a stochastic SIQR epidemic model with non-monotone incidence rate and Lévy jumps on scale-free networks is considered. When the disease will become extinct is also a concern in epidemiology, however, it is impossible to find the disease-free equilibrium of the stochastic model, to find sufficient conditions for disease extinction, we explore the asymptotic behavior of stochastic model around the equilibrium of the corresponding deterministic model.

It is not difficult to find from the conclusions of this paper that the heterogeneity of social networks accelerates the spread of diseases, that is, population shift accelerates the spread of diseases. Therefore, people should reduce mobility when there is an outbreak of disease. In addition, quarantine measures and Lévy jumps also curb the spread of diseases, the timing of quarantine warrants further study because the quarantine will have an impact on the economy. It is worth noting that there is no effect of the parameter b related to media coverage on the basic reproduction number, but it reduces the final infection scale of the disease to a certain extent. Therefore, in the initial stage of the disease outbreak, the absence of effective drug, the government can spread news about disease through various channels to enhance people's awareness of prevention, in order to reduce the threat of disease to human beings.

As we all know, the diffusion of individual does not follow a fixed pattern, but with freedom and randomness. Therefore, using local Laplacian operators to represent the spatial diffusion phenomenon cannot accurately describe the spatiotemporal behavior of the research object. The fractional Laplace operator, as a special form of nonlocal operator, can be used to describe the Lévy diffusion process and is the generator of the stable Lévy diffusion process [33]. Accordingly, on the basis of this paper, it is of great importance to introduce spatiotemporal diffusion [34], reaction–diffusion [35], shifting environment [36] and so on, in

order to establish the more realistic fractional model and spatiotemporal model. Last but not least, inspired by nonstandard numerical method [37] and evolutionary computation-based Padé approximation (EPA) scheme [38], it is of great significance to develop algorithms suitable for the model proposed in this paper to simulate the spread of diseases.

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Declaration of Competing Interest

The authors declare that there is no conflict of interest in relation to this paper.

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